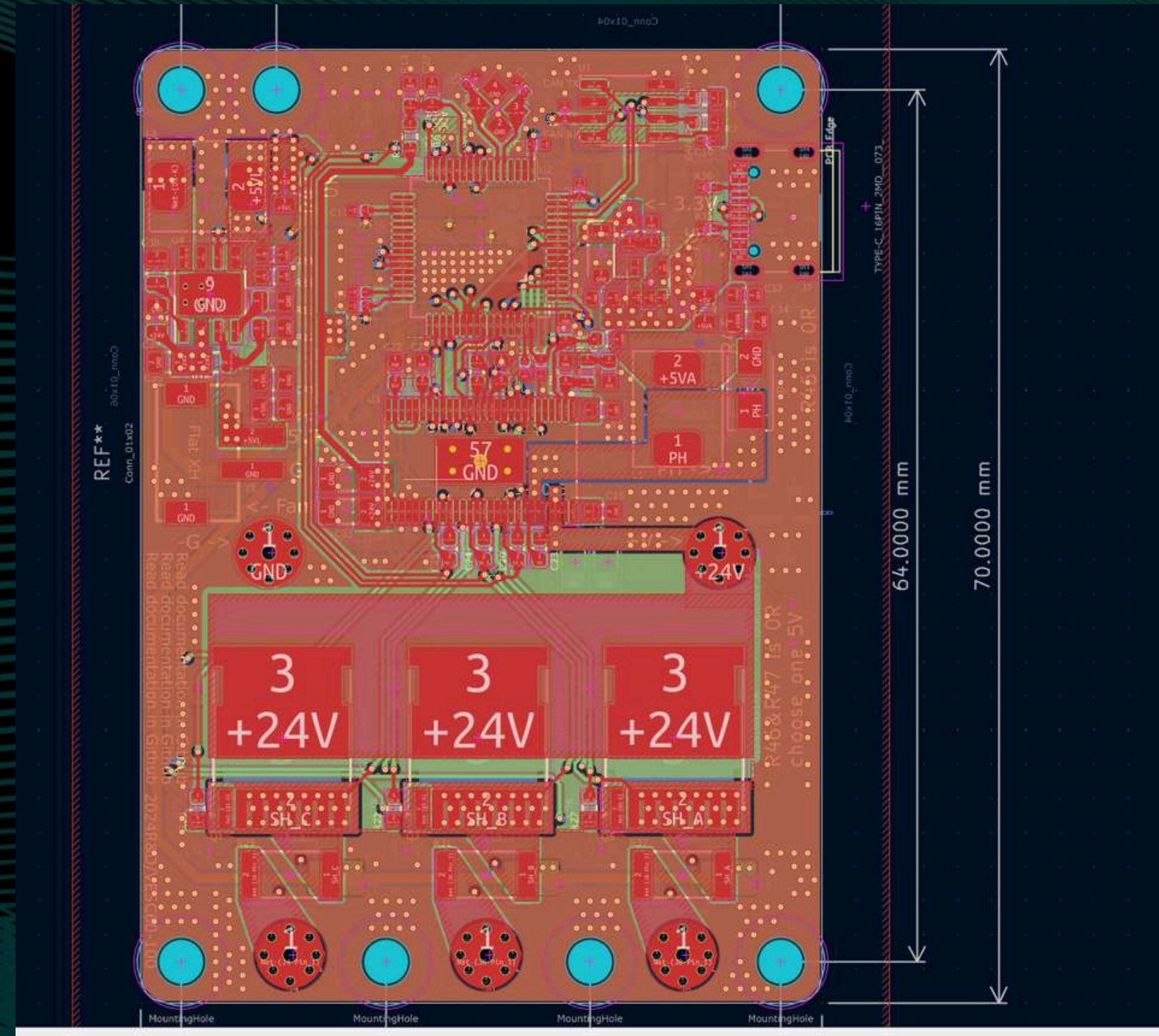
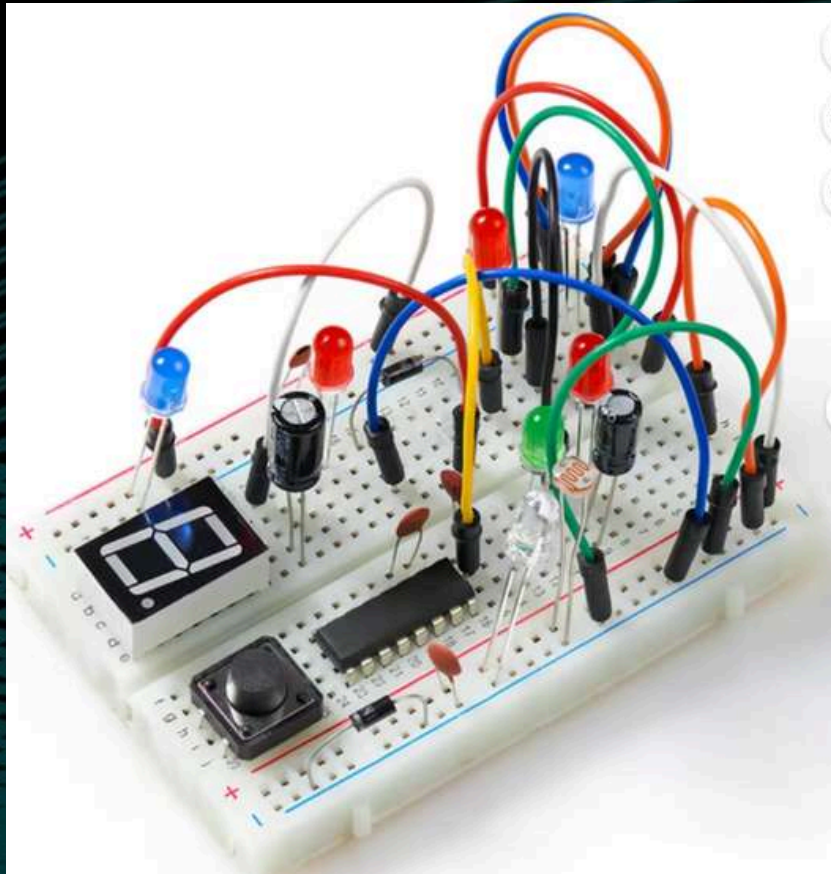


DELIVER BY Thomas (Robocon)

2026 Hardware Tutorial 2 PCB design

PCB Routing Theory

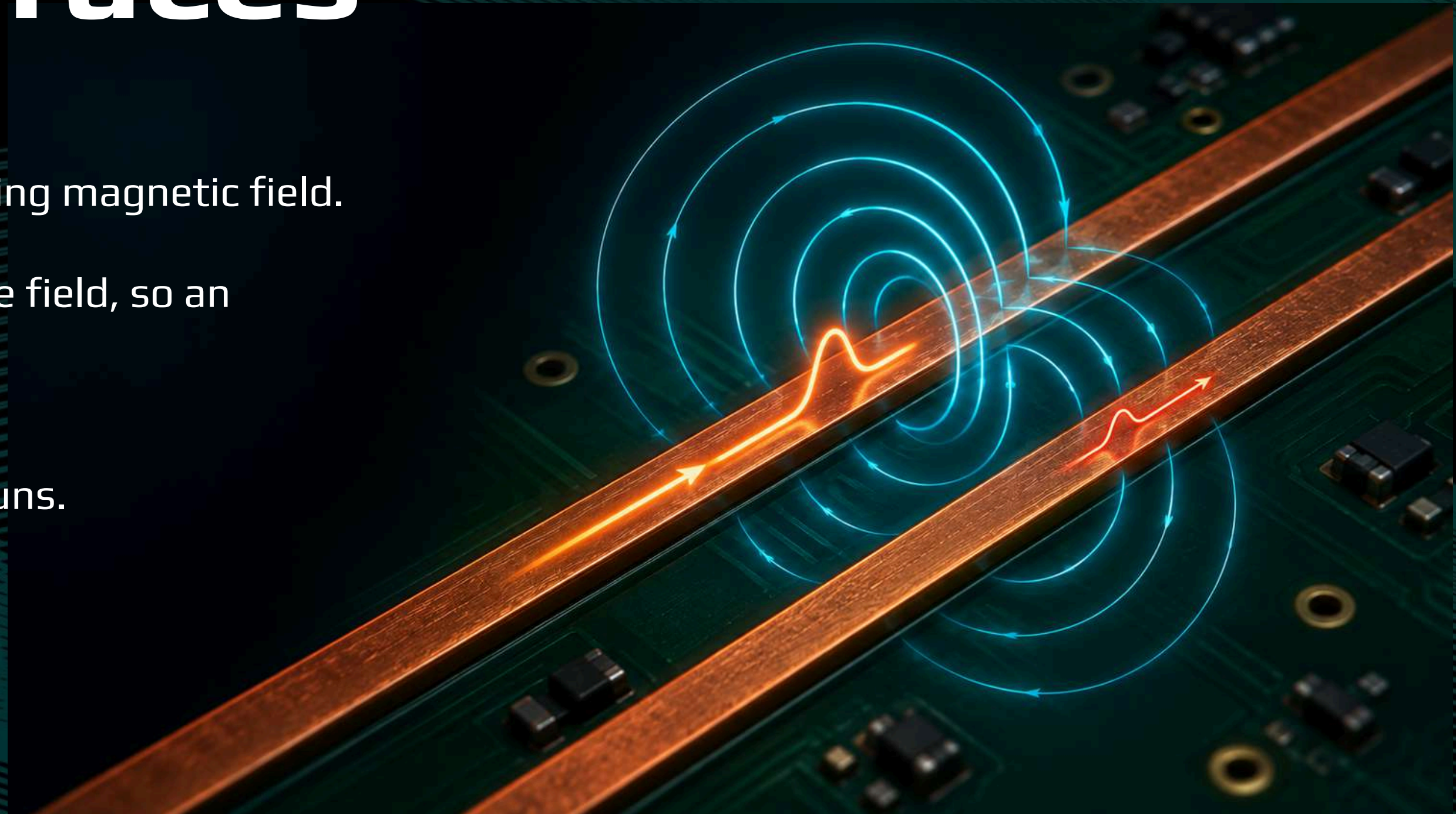


Induced Voltage Between Traces

A changing current creates a changing magnetic field.
Faraday's law: $V_{\text{ind}} = -\Delta\Phi/\Delta t$
A nearby trace intercepts part of the field, so an unwanted voltage appears.

ROUTING RULE

Increase spacing. Shorten parallel runs.
Keep the return path close.

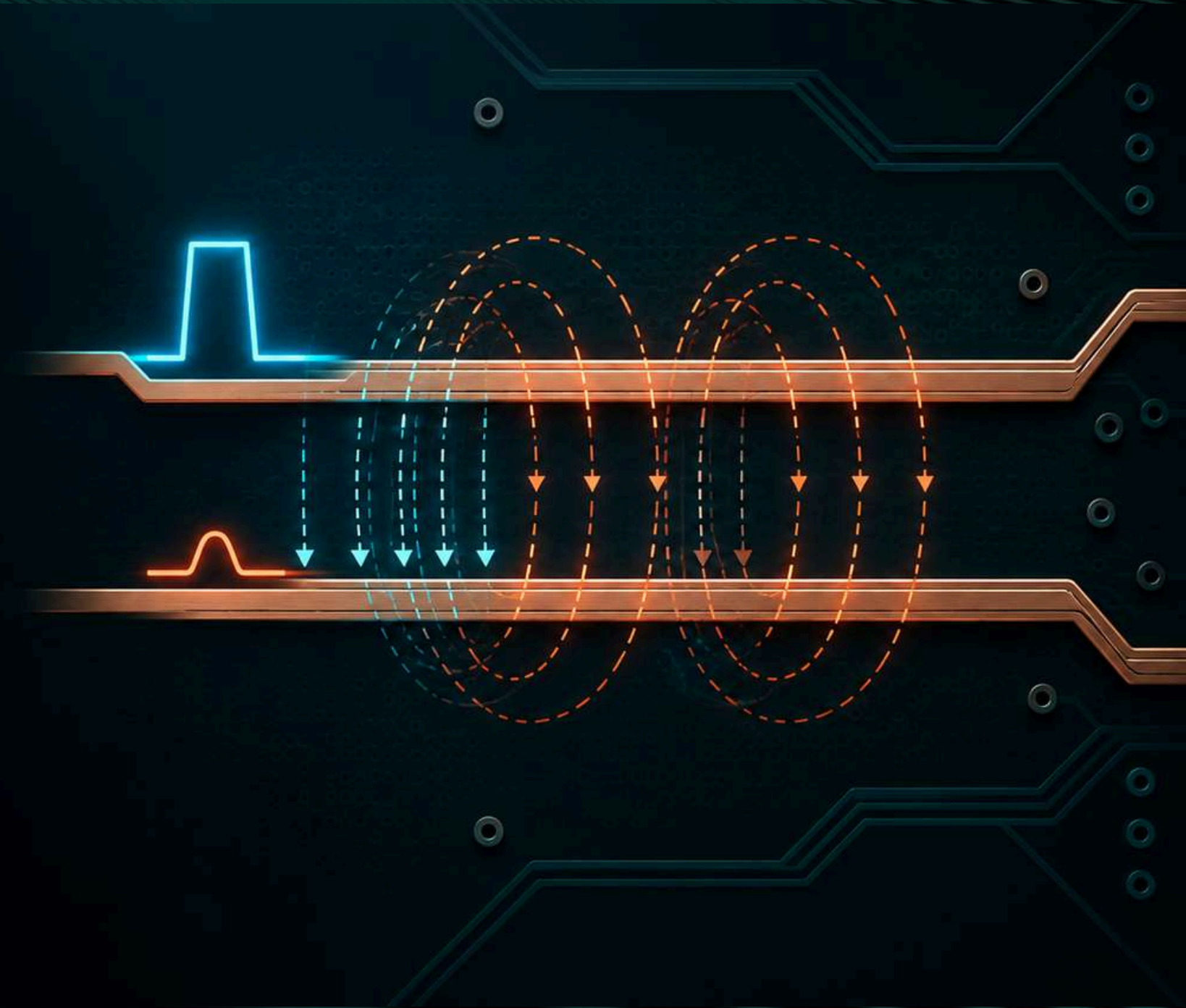


Crosstalk Between Parallel Traces

An electric field produces capacitive coupling.
A magnetic field produces inductive coupling.
Noise grows as edge speed and parallel trace length increase.

ROUTING RULE

Increase separation, keep the trace short. Reduce parallel overlap. Route above a continuous reference plane.



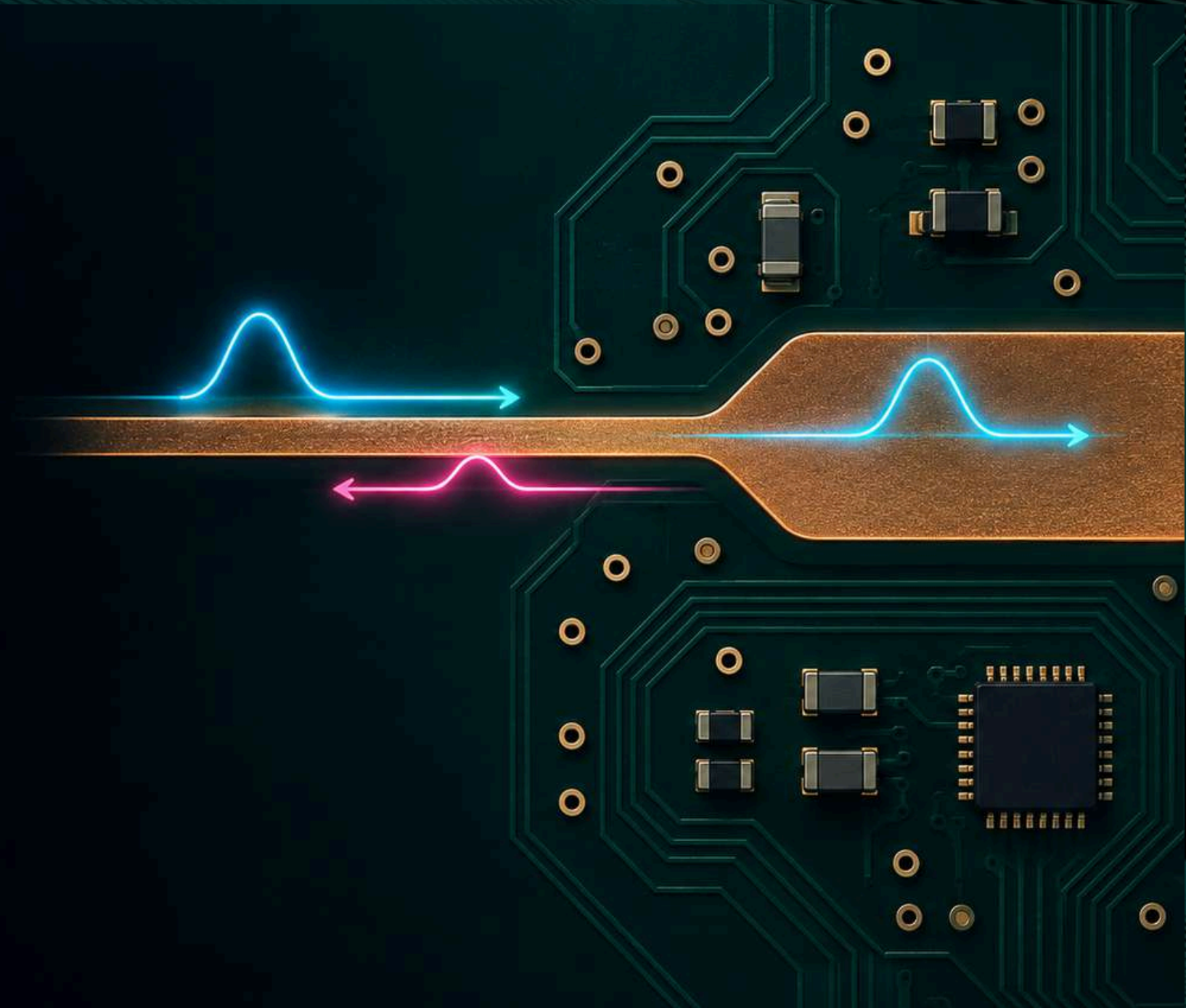
Impedance Mismatch and Reflections

A fast edge travels along the trace as a wave.
An abrupt geometry change changes the local impedance.

The reflected energy causes ringing and overshoot.

ROUTING RULE

Keep trace width and the reference plane consistent.
Avoid stubs and sudden transitions.

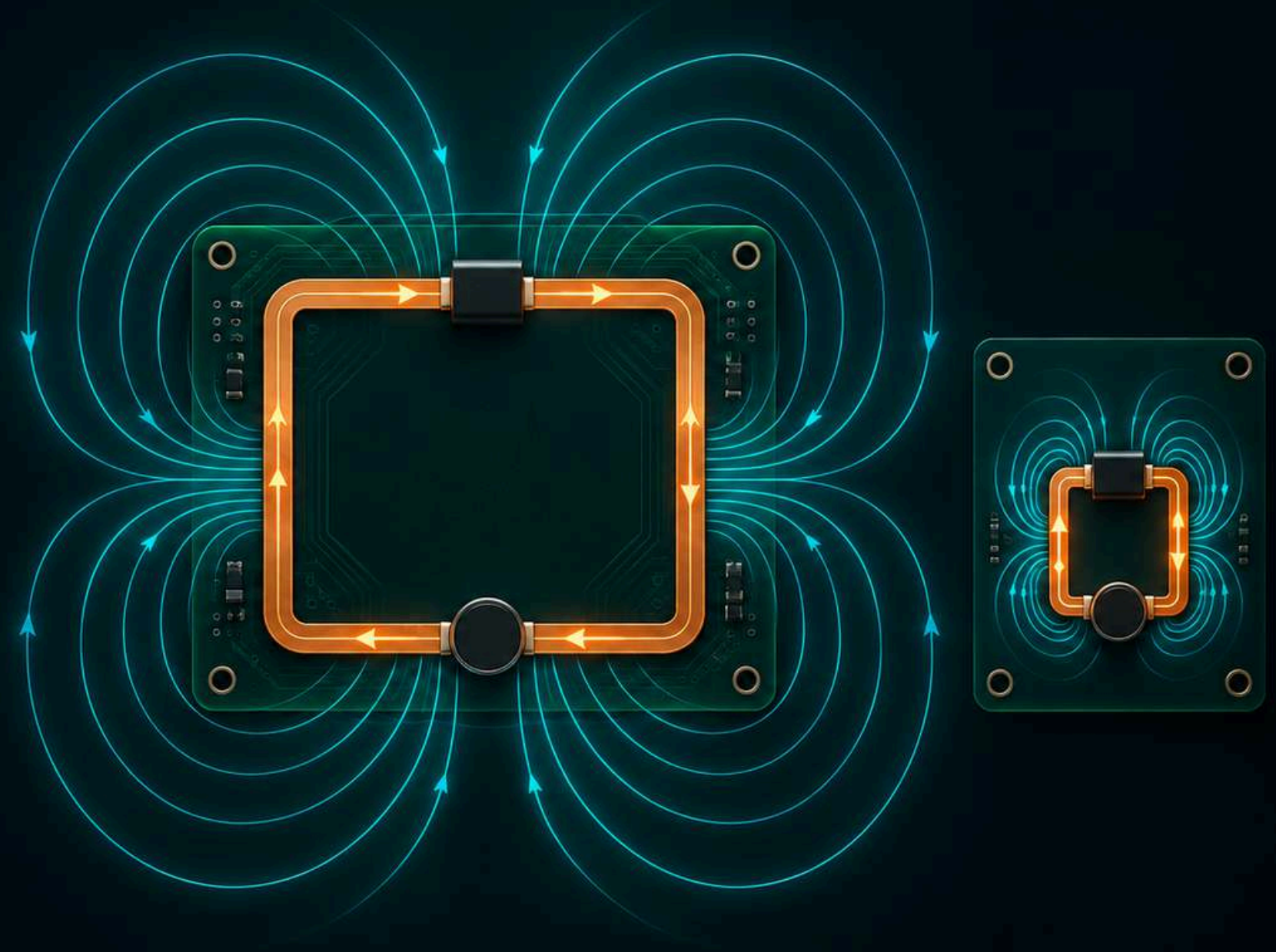


Loop Area Controls EMI

Current flows around a complete loop through the signal and its return path.
A larger loop encloses more magnetic flux and radiates more strongly.
Changing flux can also induce noise in nearby circuits.

ROUTING RULE

Keep signal and return paths close. Use a continuous ground plane. Avoid plane gaps.



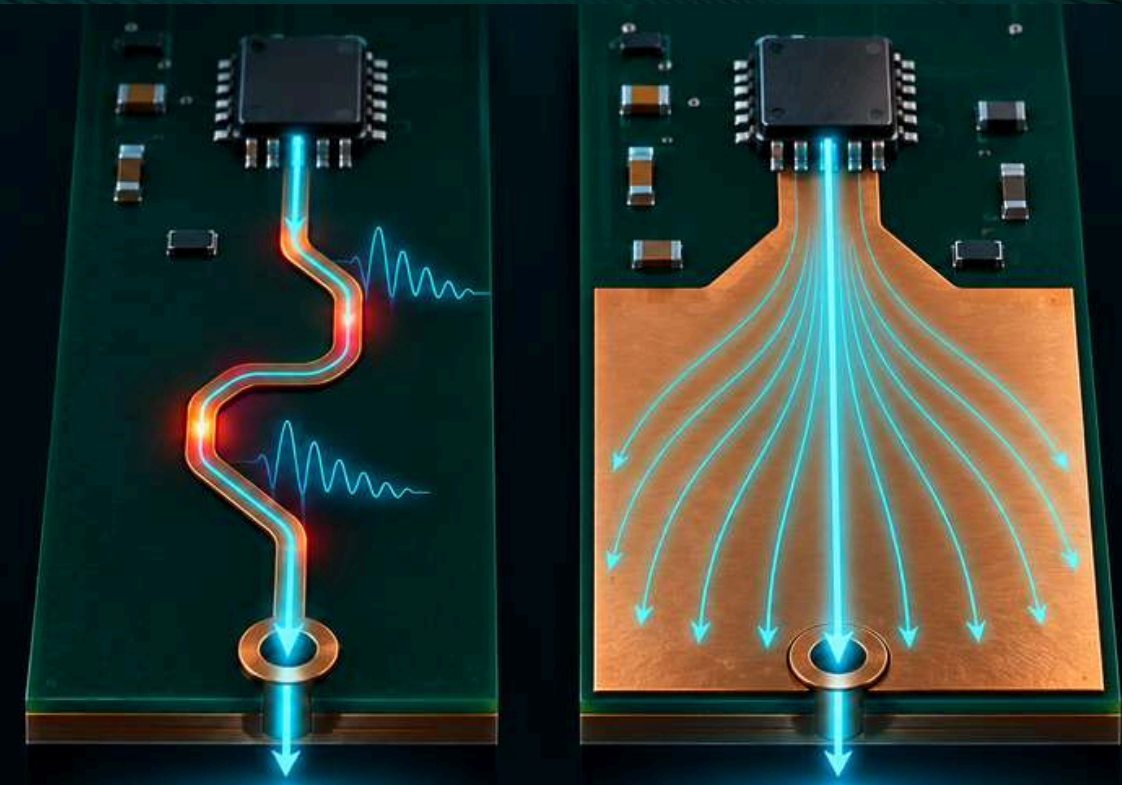
Ground Plane = Low Impedance

PHYSICS

Ground noise follows $V_{noise} = IR + L(di/dt)$. A wide copper plane has lower resistance and inductance than a narrow trace. It spreads return current and keeps the reference voltage stable.

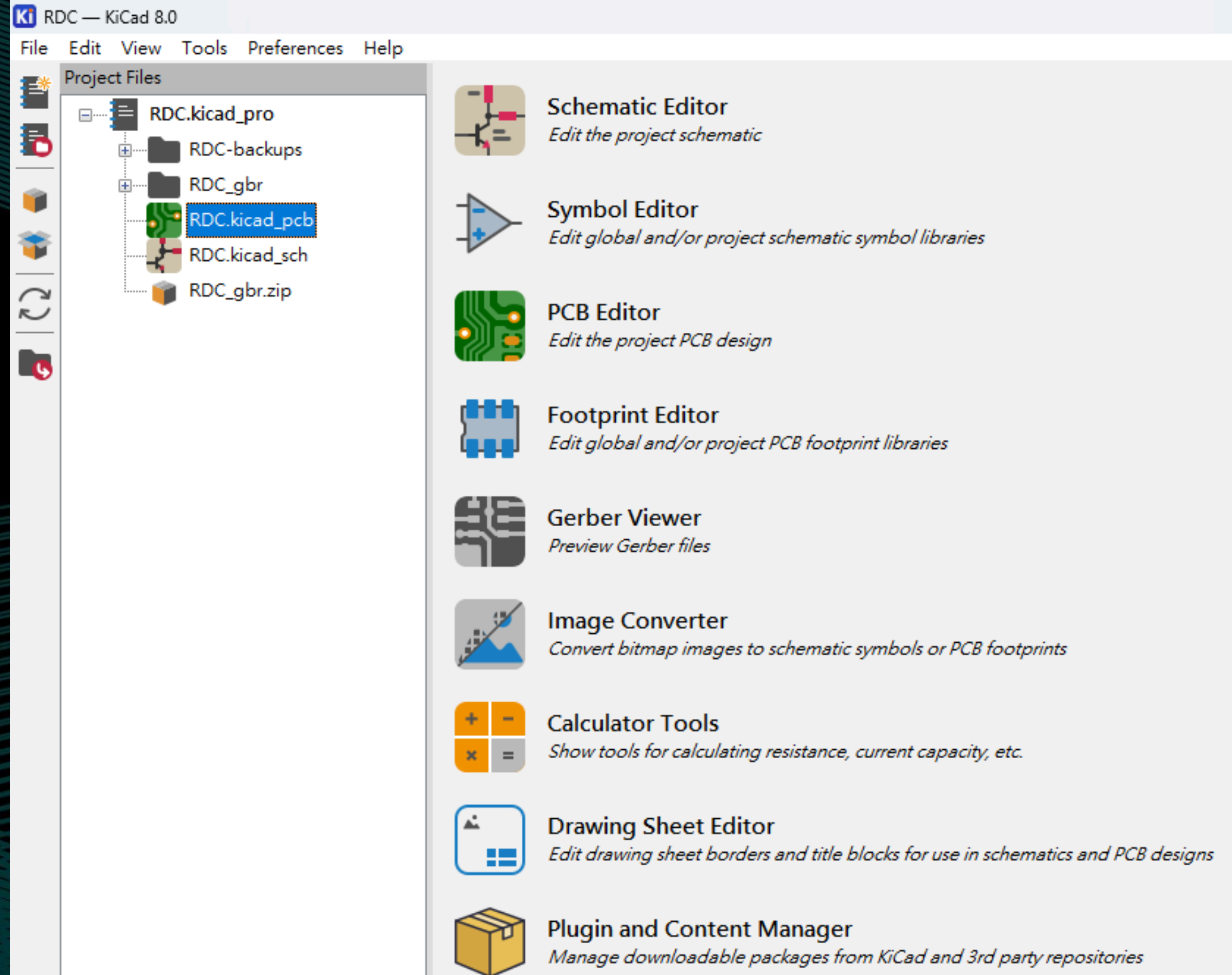
ROUTING RULE

Use an unbroken ground plane. Keep high-current returns away from sensitive analog paths.

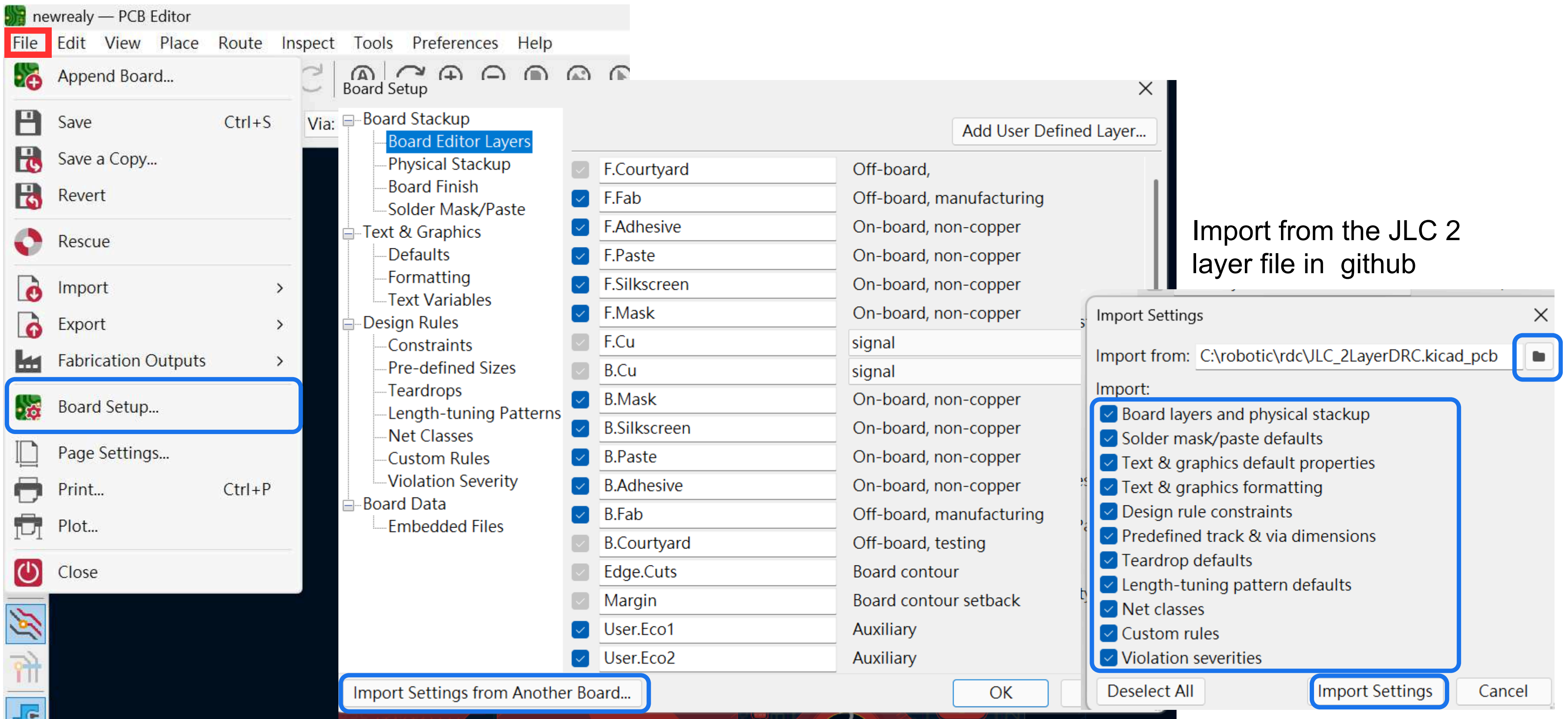


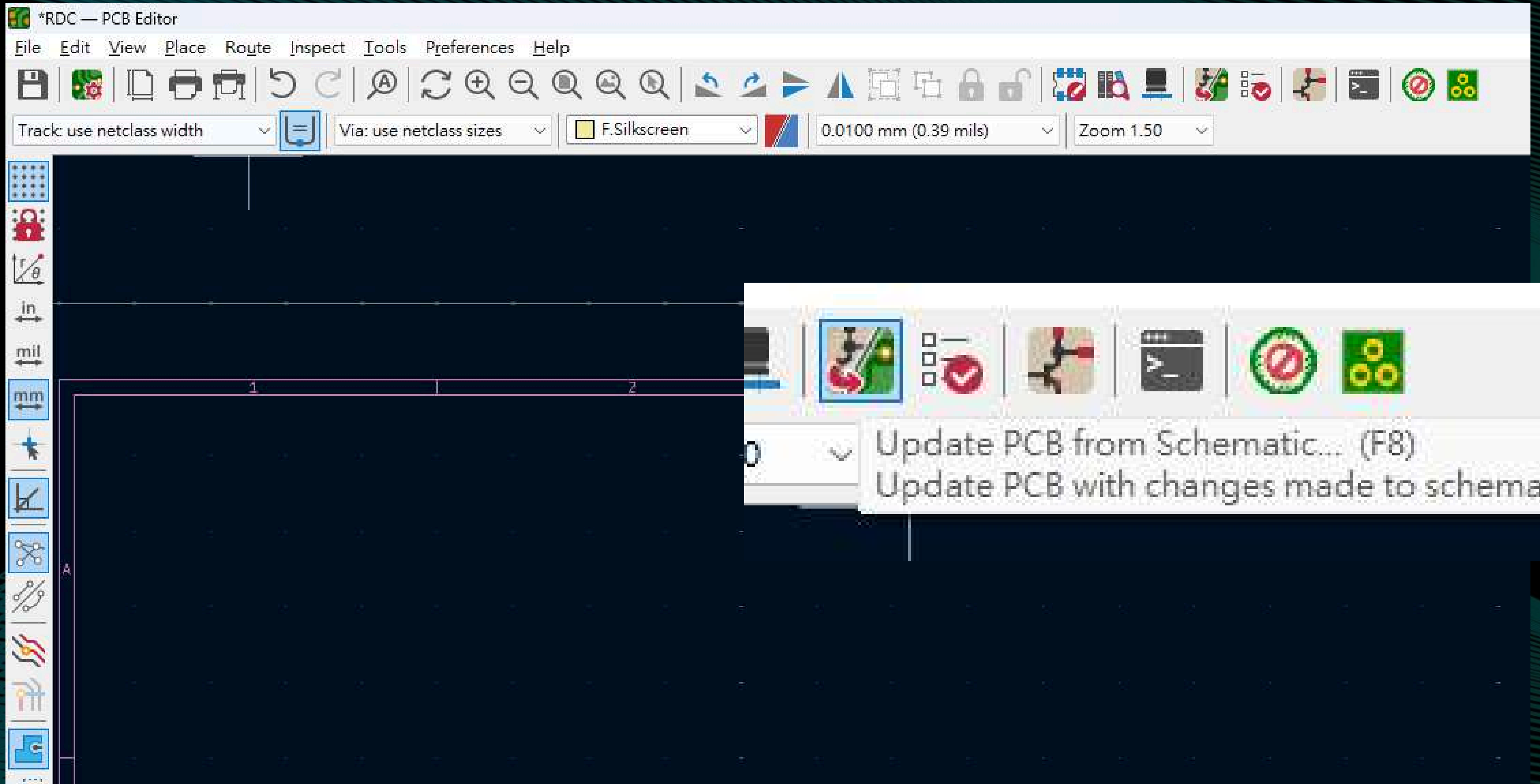
PCB

STEP OF SCH TO PCB



Import DRC setting





Update PCB from Schematic



Options

- ☐ Re-link footprints to schematic symbols based on their reference designators
- ☐ Delete footprints with no symbols
- ☒ Replace footprints with those specified in the schematic

Changes To Be Applied

Add U4 (footprint Package_SO:SOIC-8_3.9x4.9mm_P1.27mm).
Add J9 (footprint Connector_PinHeader_2.54mm:PinHeader_1x08_P2.54mm_Vertical).
Add U5 (footprint Package_SO:SOIC-8_3.9x4.9mm_P1.27mm).
Add J10 (footprint Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical).
Add U6 (footprint Package_SO:TI_SO-PowerPAD-8_ThermalVias).
Add J11 (footprint Connector_PinSocket_2.54mm:PinSocket_2x03_P2.54mm_Vertical).
Add U7 (footprint HKUST_RDC2024_Footprint_Lib:SOT-223).
Add J12 (footprint Connector_PinSocket_2.54mm:PinSocket_2x03_P2.54mm_Vertical).
Add U8 (footprint HKUST_RDC2024_Footprint_Lib:SOT-223).
Add J13 (footprint Connector_PinSocket_2.54mm:PinSocket_2x03_P2.54mm_Vertical).
Add Y1 (footprint HKUST_RDC2024_Footprint_Lib:Crystal_HC49-4H_Vertical).

Total warnings: 0, errors: 0.

Show: ☒ All ☒ Errors 0 ☒ Warnings 0 ☒ Actions ☒ Infos

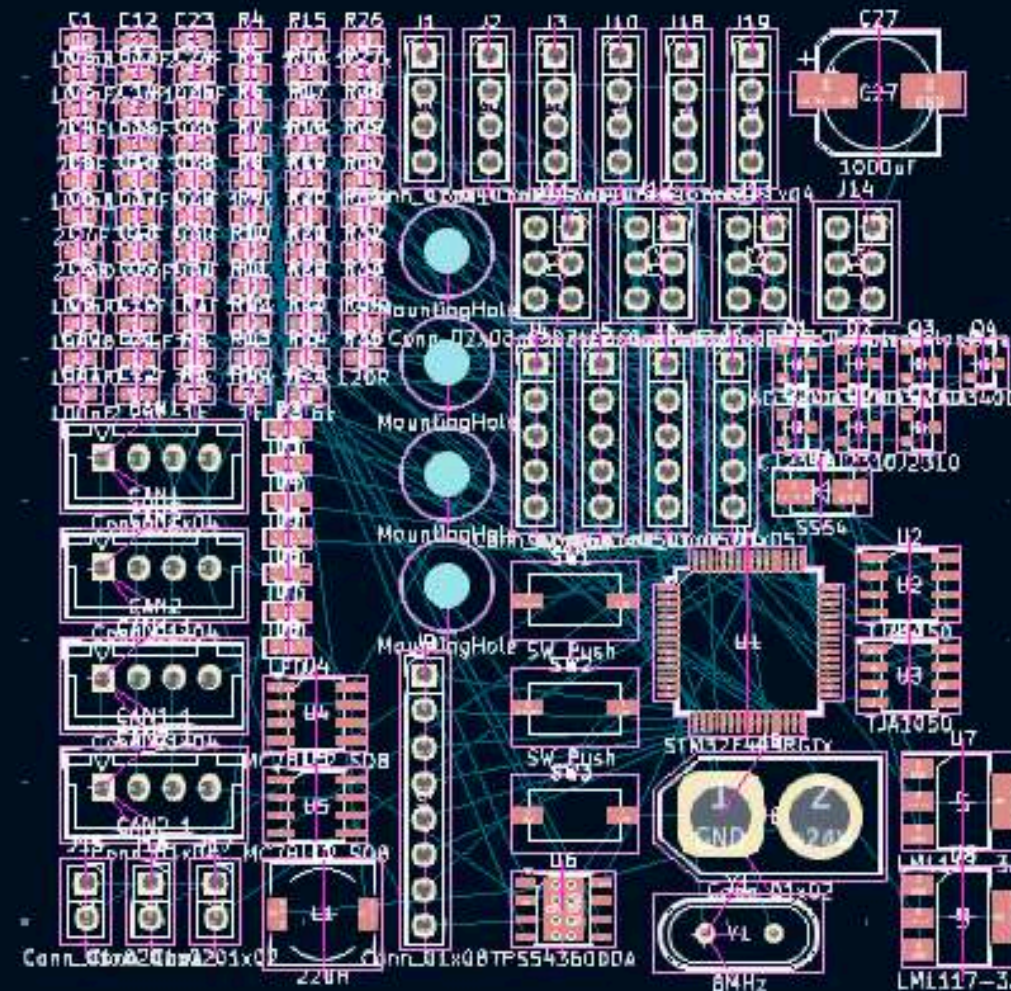
Save...

Update PCB

Close

Step 1: Plan before you route

wire will block other wire so you will be advised to plan the routing first

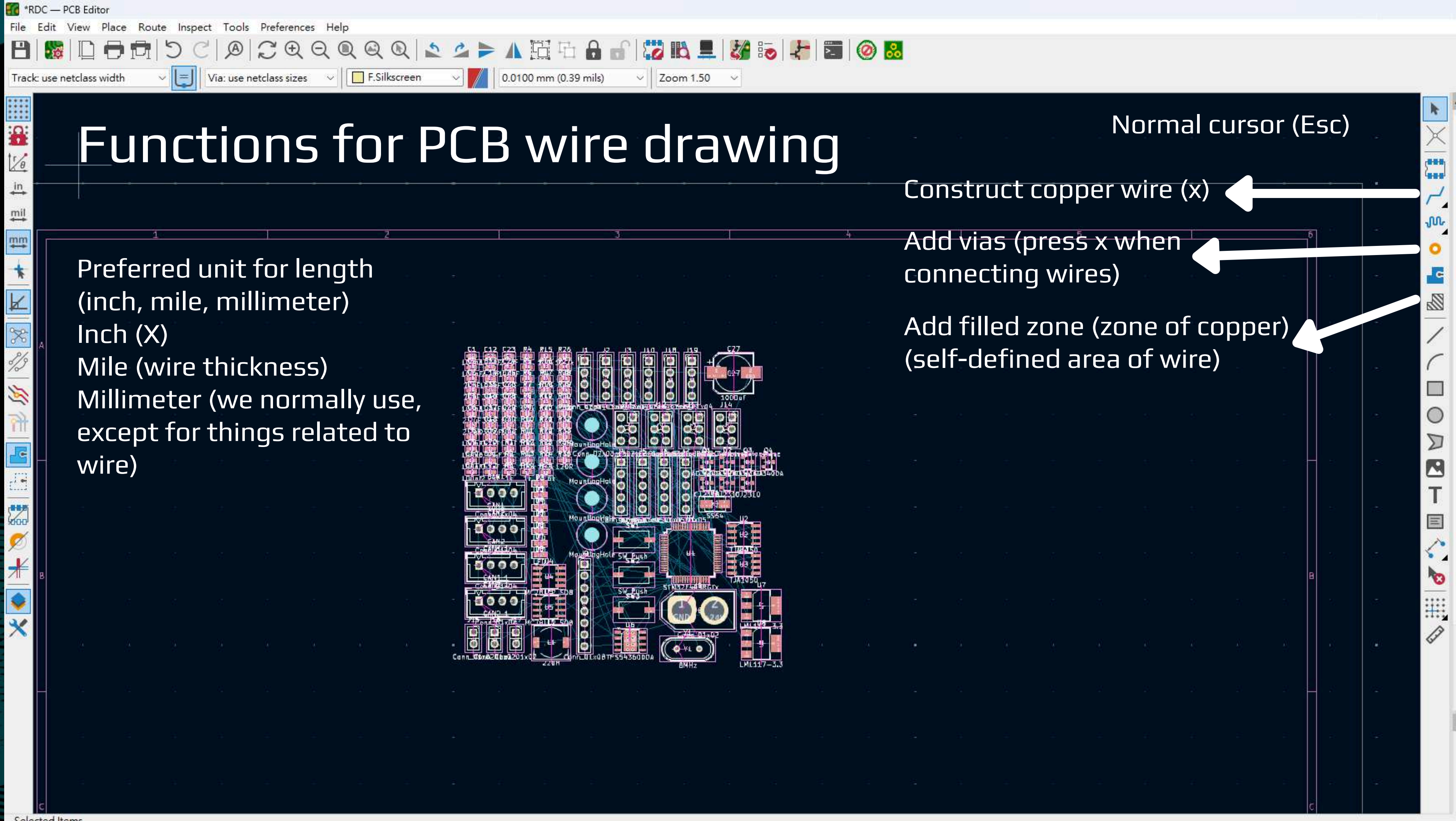


Suggested Routing Priority

1. high-speed signal
2. power
3. GPIO (led, button etc)

How to use Kicad to draw PCB

LETS TURN THE SCHEMATIC INTO A PCB!!!

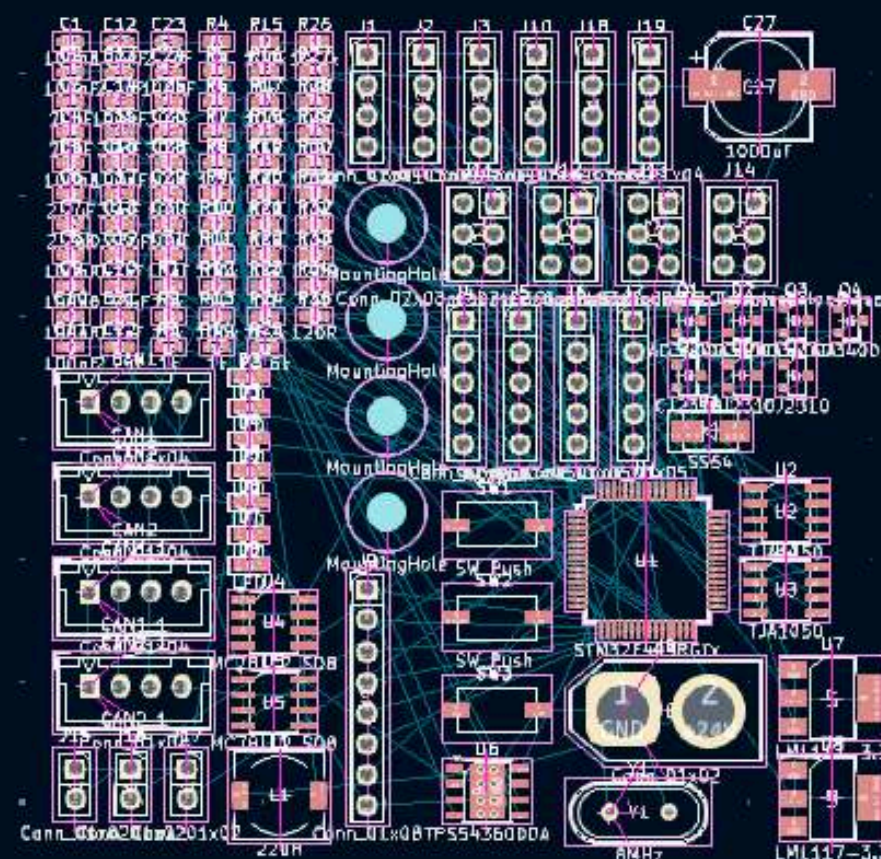




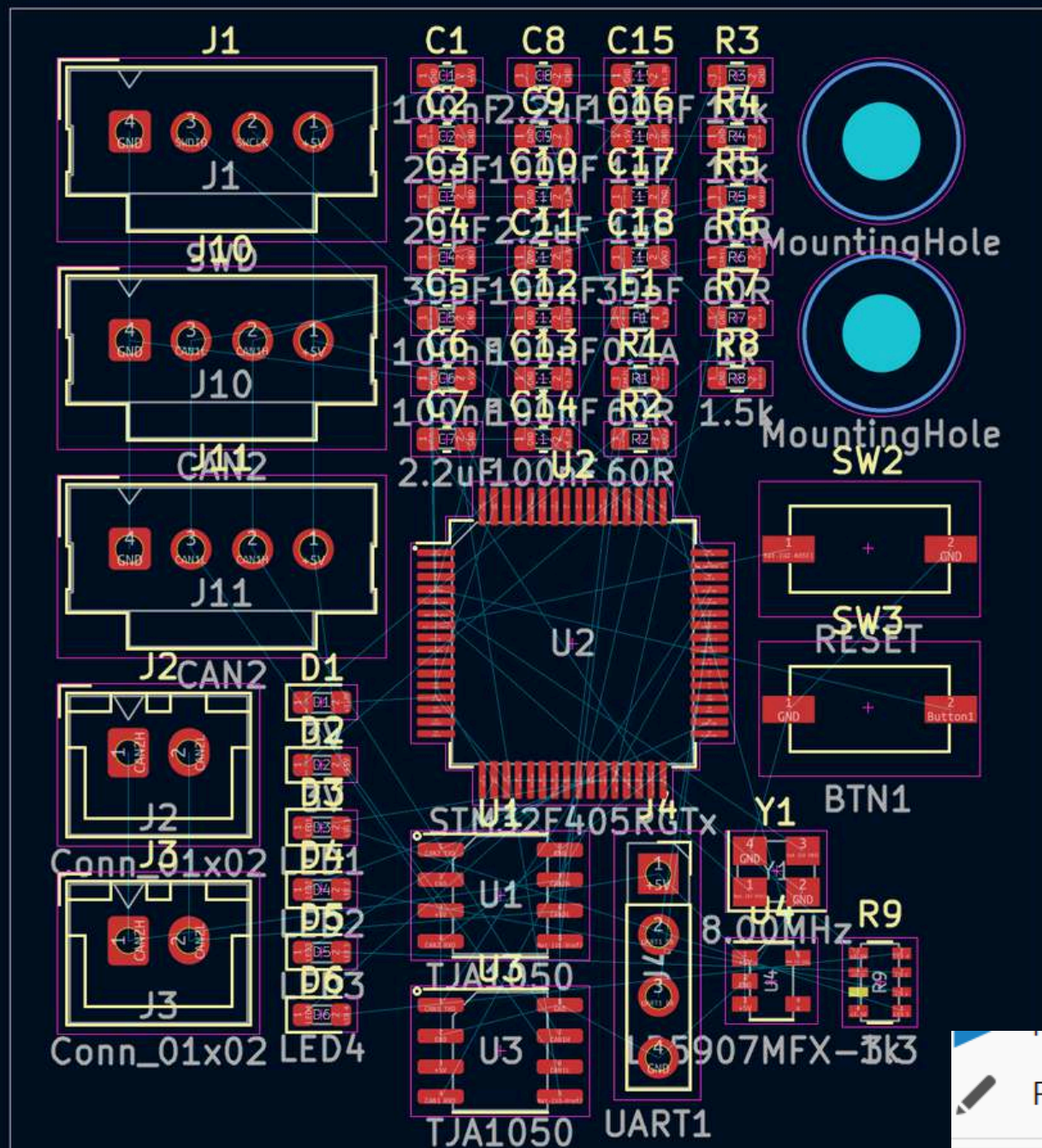
Track: use netclass width Via: use netclass sizes F.Silkscreen 0.0100 mm (0.39 mils) Zoom 1.50

Functions for edge cut drawing / user drawing

Draw line
Arc
Rectangle
Circle
Polygon



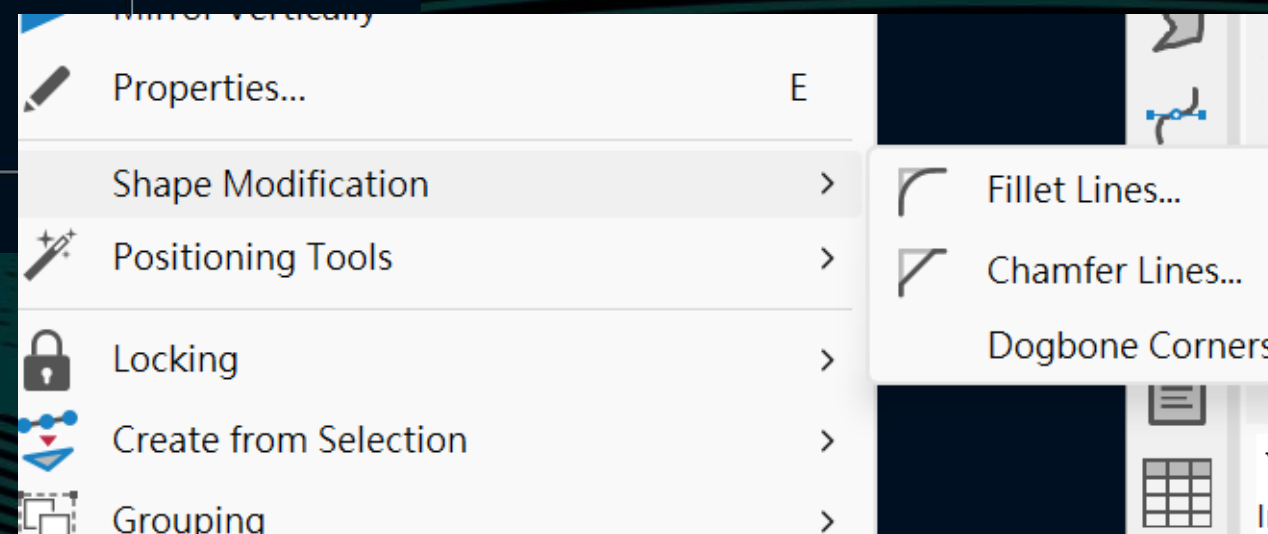
Step 2: Make an edge cut (the board edge)



Draw a rectangle in edge cut layer covering all of the components, you should estimate how big you board is

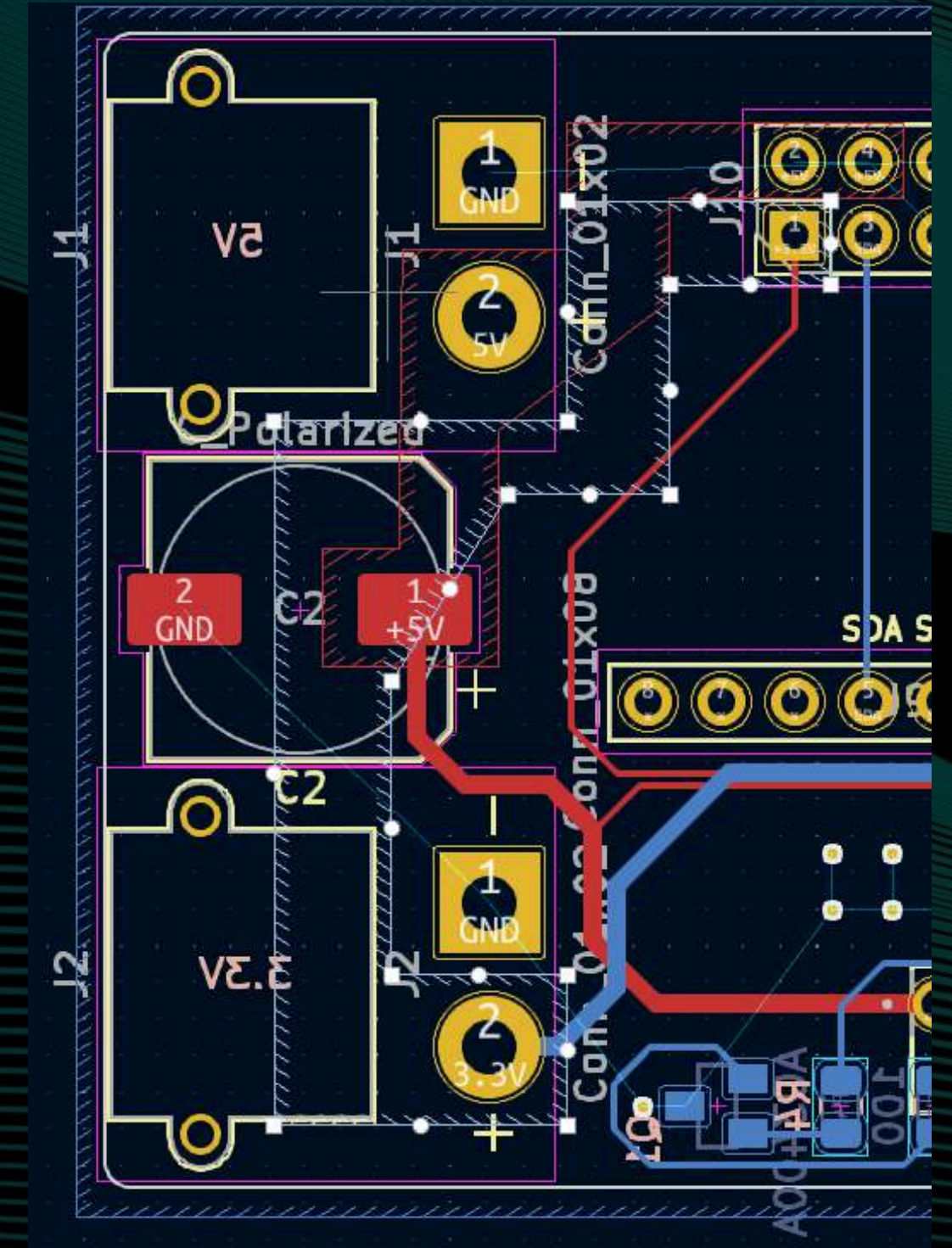
Note for RDC:

It should be smaller than 10cm x 10cm!!!



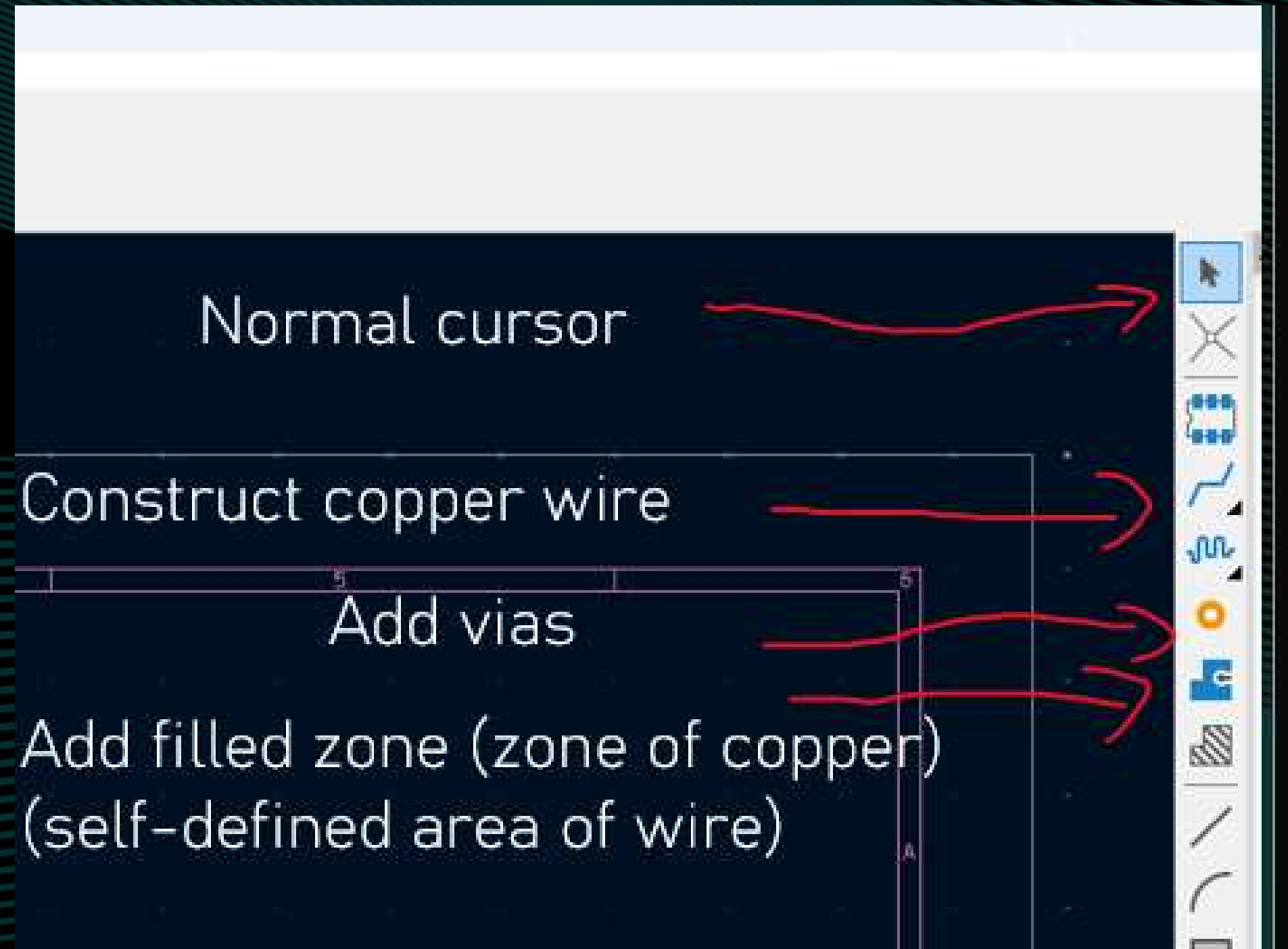
Zone

is a zone of copper
You can imagine it is a wire/track that you can
customize the shape and fill



Zone

How to add zone



How to add zone

If you want to add zone for net that you cannot find
Make sure you untick this box first

The screenshot shows the 'Copper Zone Properties' dialog box. It has several sections: 'Layer', 'Net', 'General', 'Electrical Properties', and 'Fill'. Annotations with arrows point to specific elements: 'Layer to fill' points to the 'F.Cu' and 'B.Cu' checkboxes; 'What to fill (usually GND / Power)' points to the 'Net' list where '+3.3V' is selected; 'Higher priority means it will fill first when there is an overlap with a lower-priority zone' points to the 'Zone priority level' dropdown set to '3'; 'Use solid for better connection' and 'Use thermal relief for better heat dissipation' point to the 'Pad connections' dropdown set to 'Solid'; and another arrow points to the 'Hide automatically generated net names' checkbox, which is currently unchecked.

Copper Zone Properties

Layer

- ☒ F.Cu
- ☒ B.Cu

Net

Filter

- <no net>
- +24V
- +3.3V**
- +5V
- /Button1
- /Button2
- /CAN1H
- /CAN1L
- /CAN1_RX
- /CAN1_TX
- /CAN2H
- /CAN2L

☐ Hide automatically generated net names ☐ Sort nets by pad count

General

Zone name:

Zone priority level: 3

Shape

☐ Locked

Outline display: Hatched

Outline hatch pitch: 19.68503937 mils

Corner smoothing: None

Fillet radius: 0 mils

Electrical Properties

Clearance: 8 mils

Minimum width: 10 mils

Pad connections: Solid

Thermal relief gap: 19.68503937 mils

Thermal spoke width: 19.68503937 mils

Fill

Fill type: Solid fill

Orientation: 0°

Hatch width: 39.37007874 mils

Hatch gap: 59.05511811 mils

Smoothing effort: 0

Smoothing amount: 0.10

Remove islands: Always

Minimum island size: 15500.031 mils²

OK Cancel

Layer to fill

What to fill
(usually GND / Power)

Higher priority means
it will fill first when
there is an overlap with
a lower-priority zone

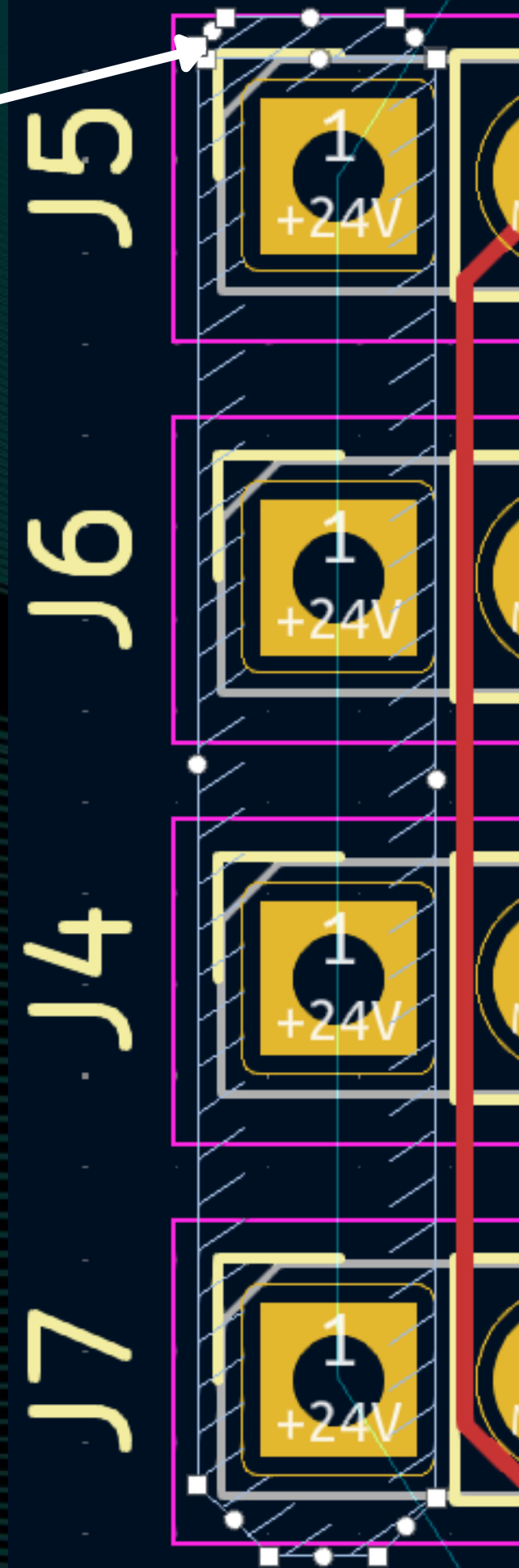
Usually:

- 0 for GND
- 1 for 12V
- 2 for 5V
- 3 for 3.3V

Use solid for better connection
Use thermal relief for better heat
dissipation

How to add zone

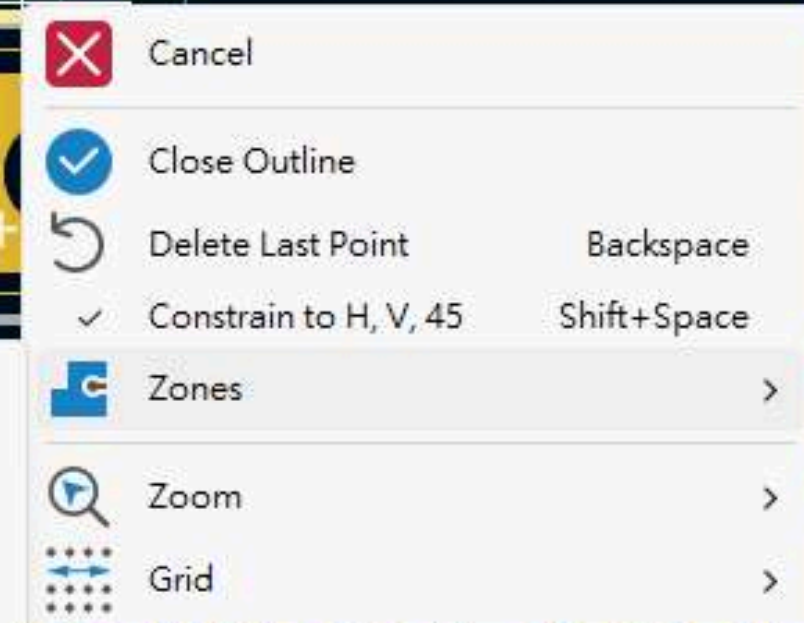
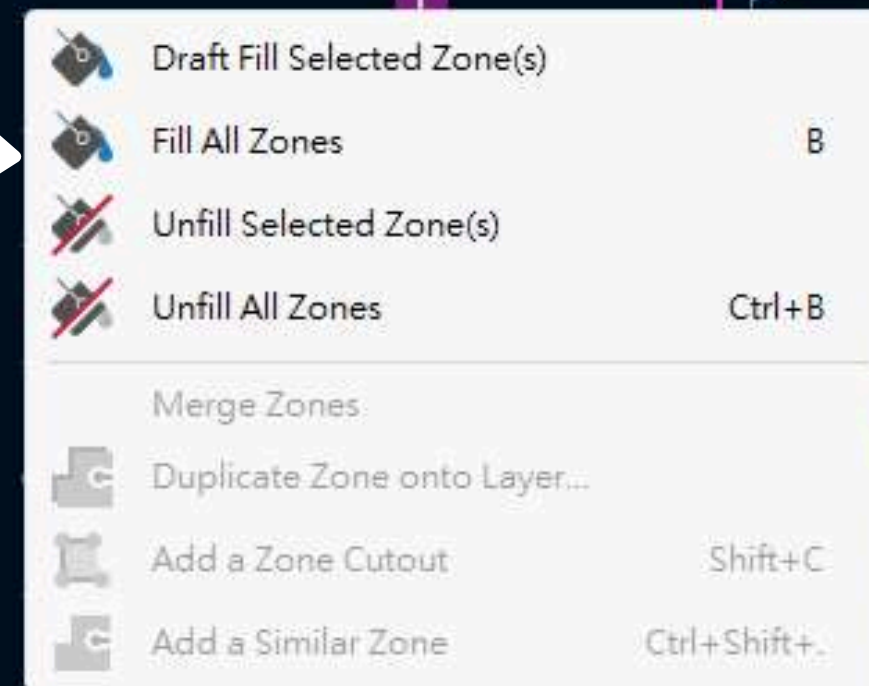
Draw with 45 degree corner instead of 90 degree



How to add zone

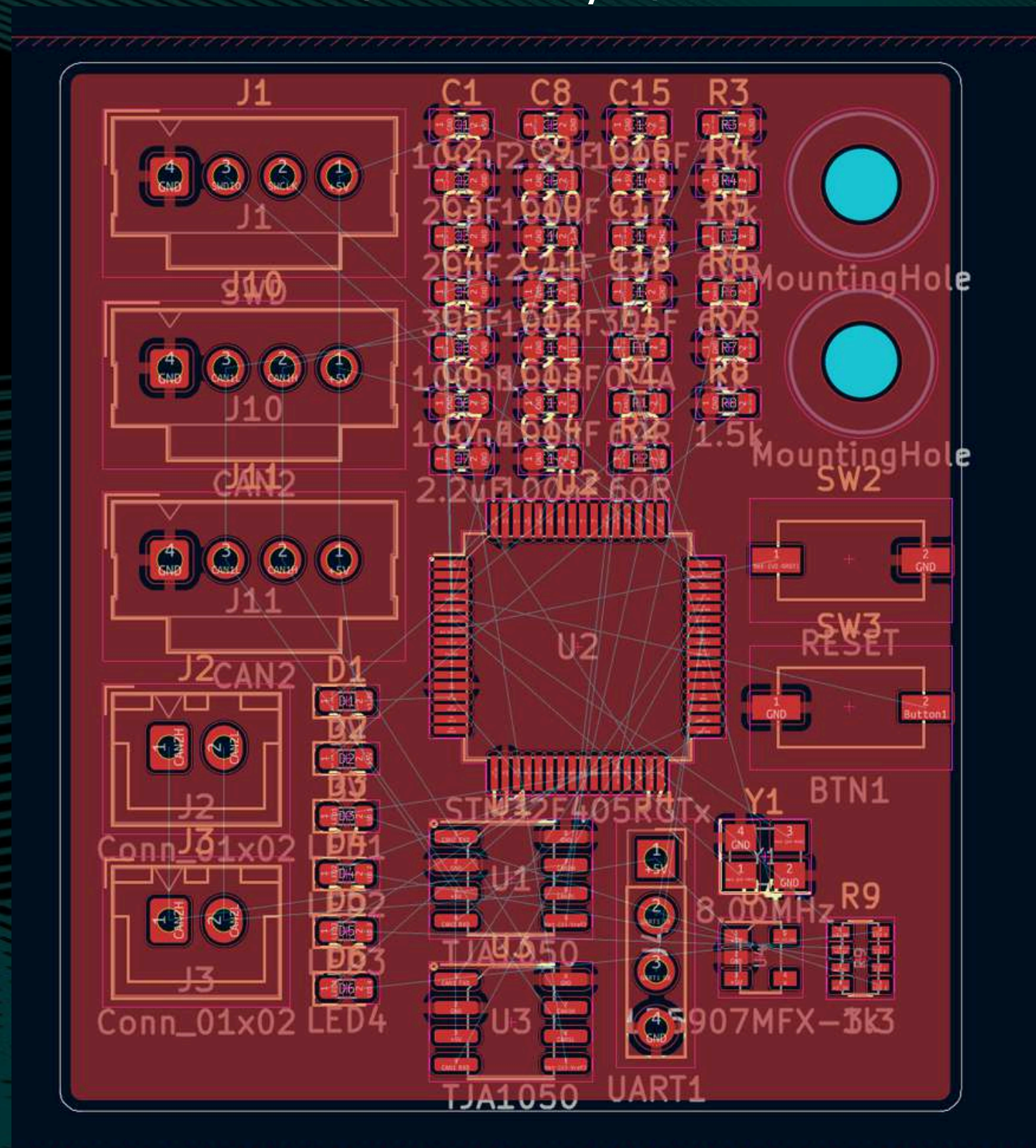
Press b on the key board to fill

Or press this to fill

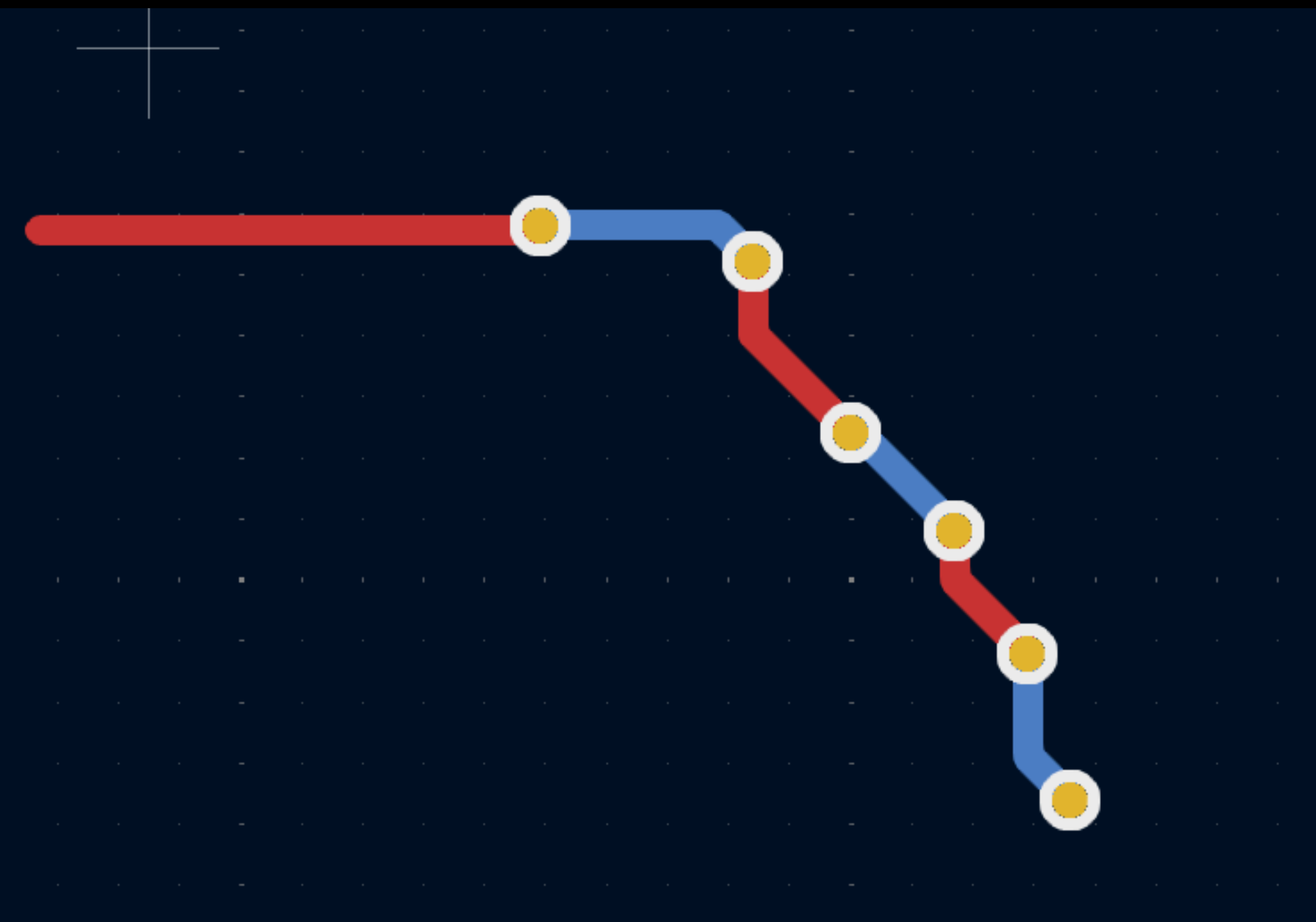


How to add zone

add a gnd zone for the whole board (f and b layer)



Key point you need to prevent:
Too many vias for one track
90 degree / right angle track routing

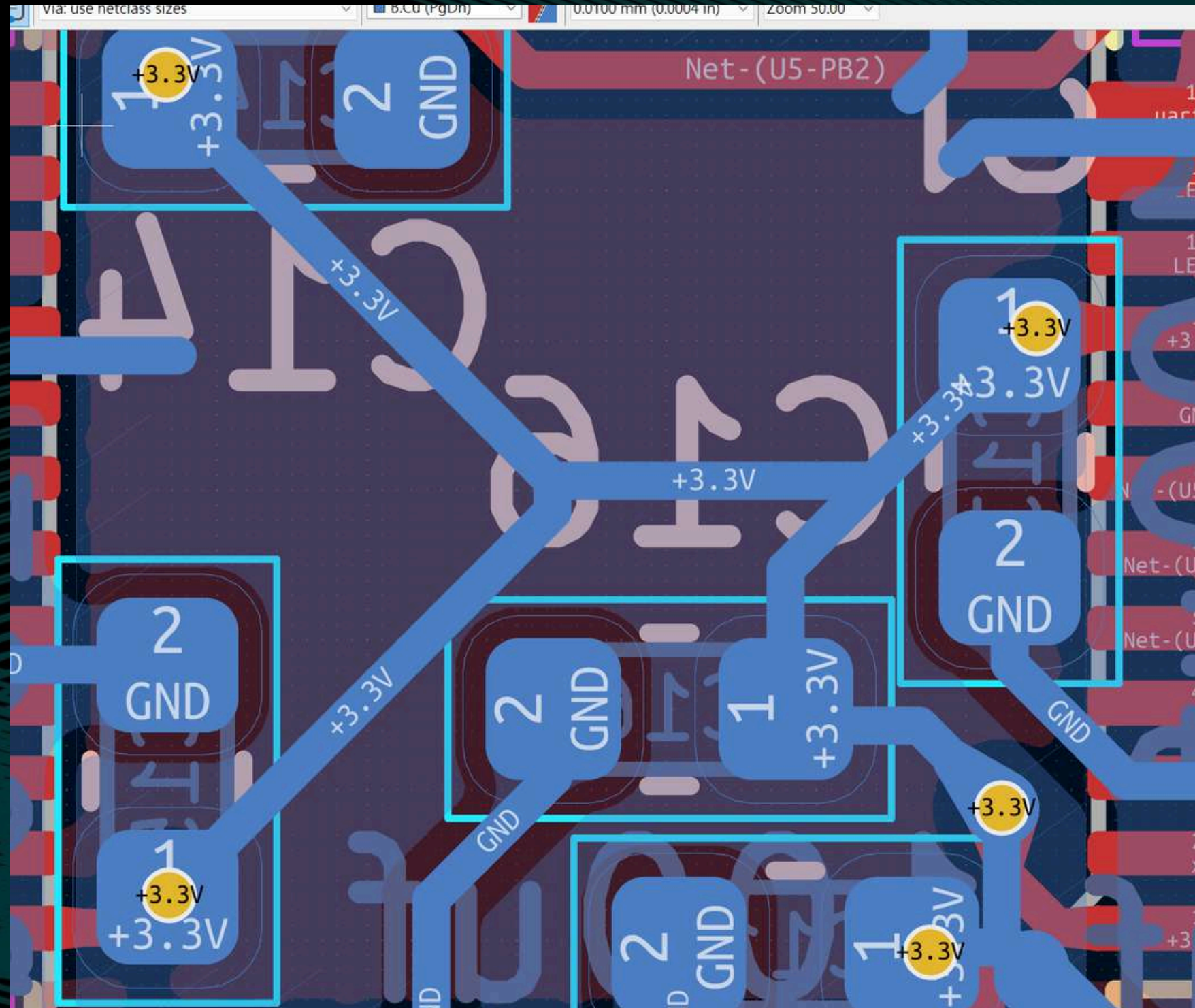


Key point you need to prevent:
Don't put any track near edge cut especially power track

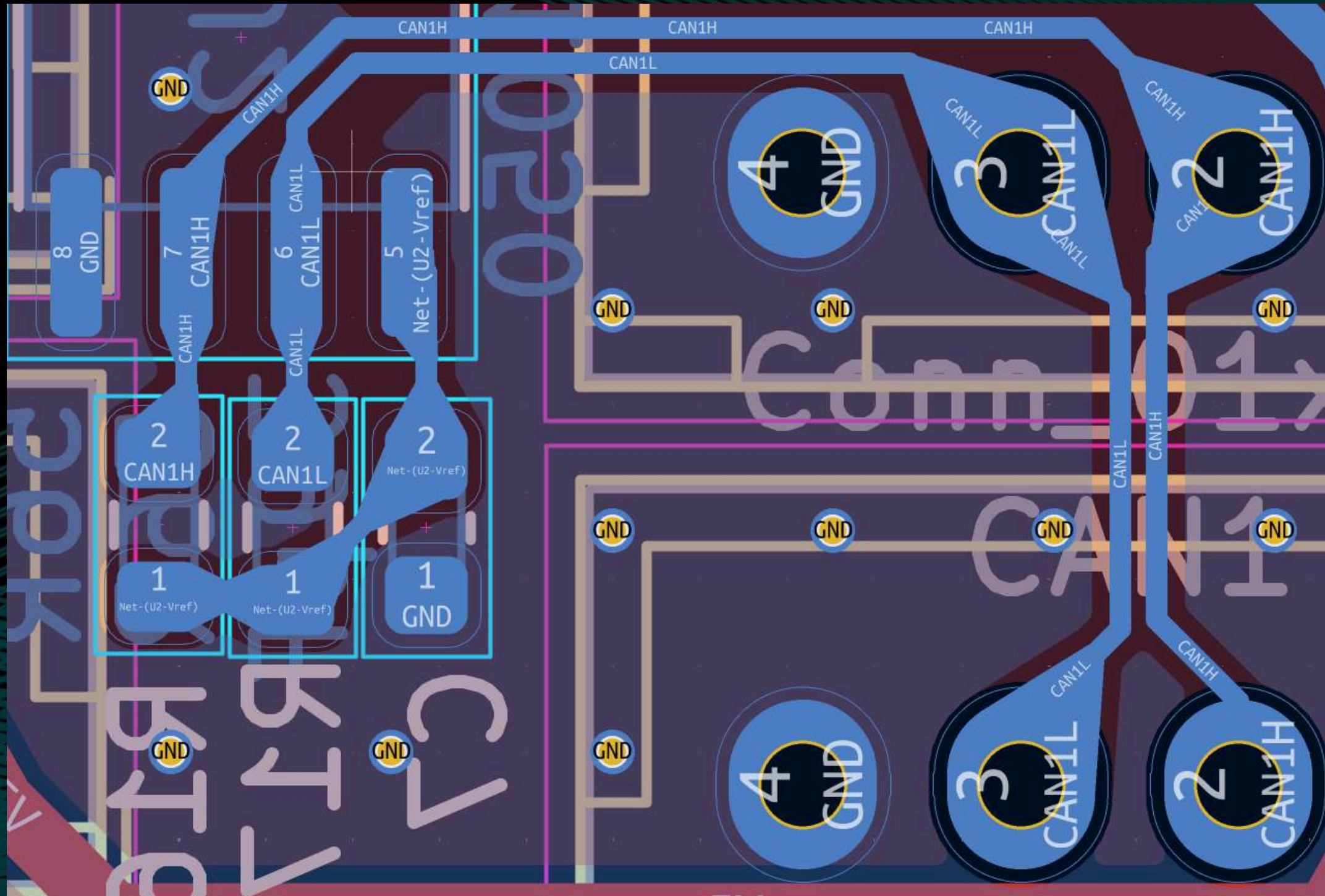


+5V

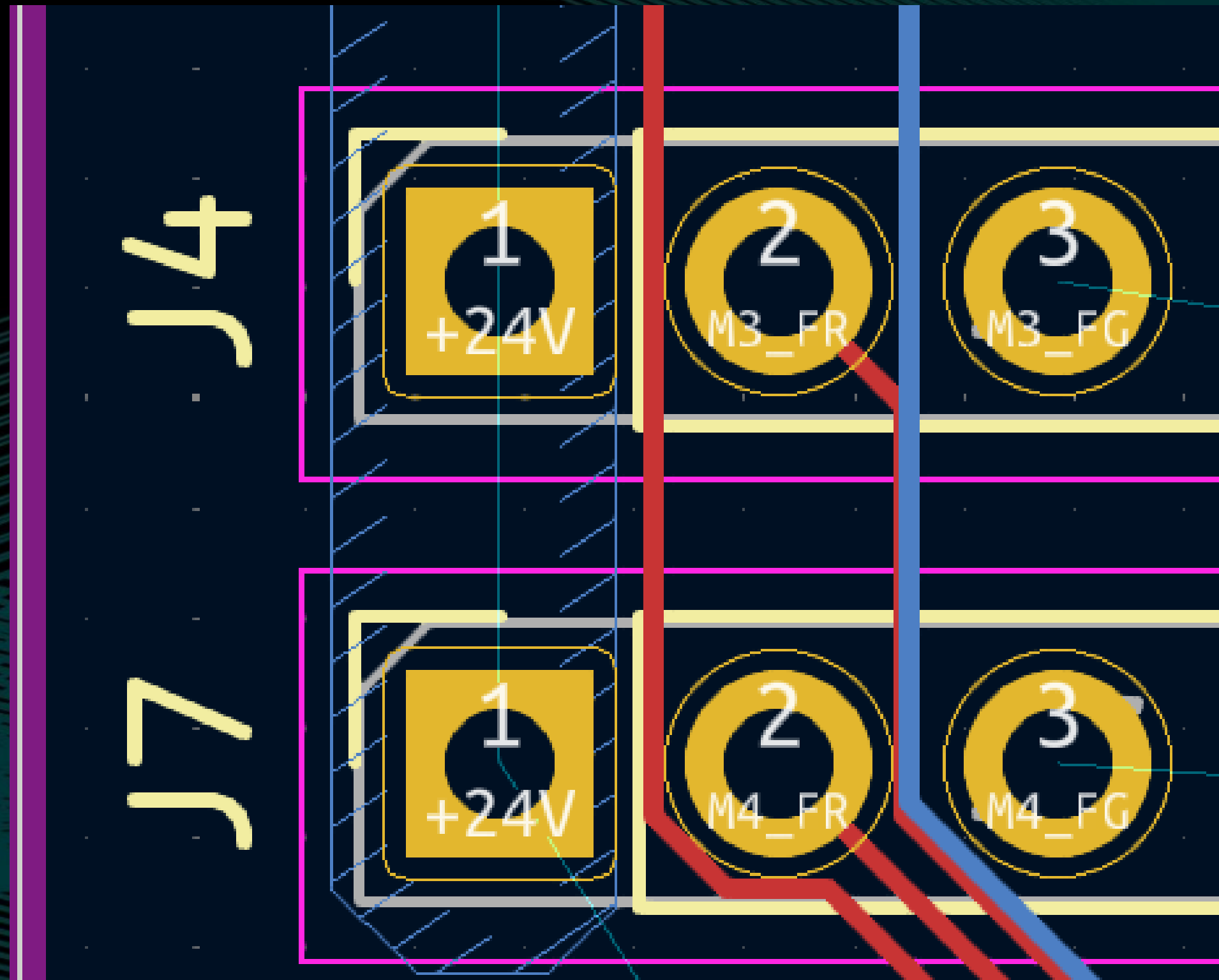
Key point you need to prevent:
too messy track



Important notes:
differential pair signals, they should be having same impedance i.e same length



Key point you need to prevent:
Overlap track for high frequency track (just try to prevent too many overlaps in your routing)
Try to prevent the route track between two connectors
Try to prevent the route track is just next to the through hole

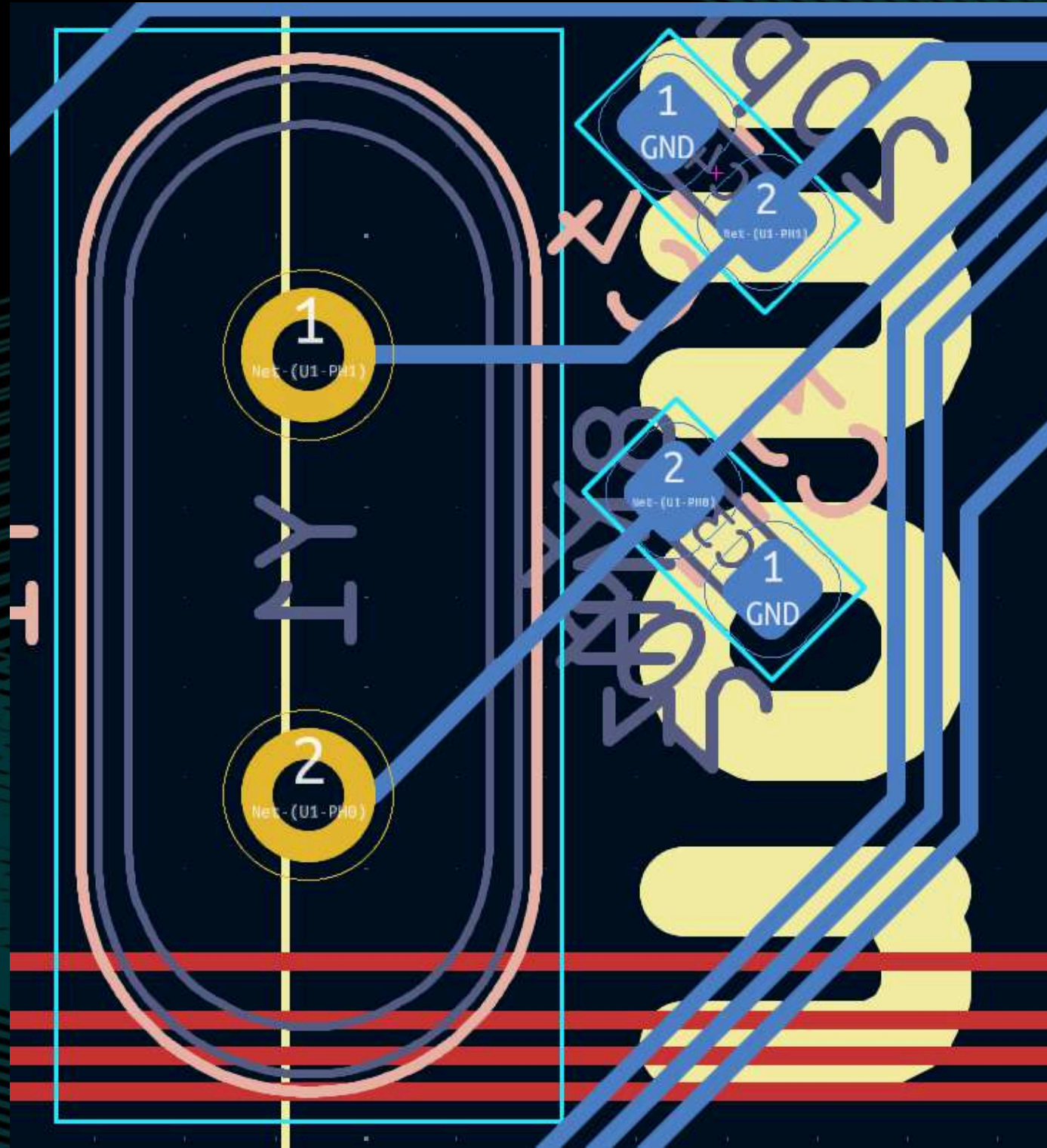


Key point you need to prevent:

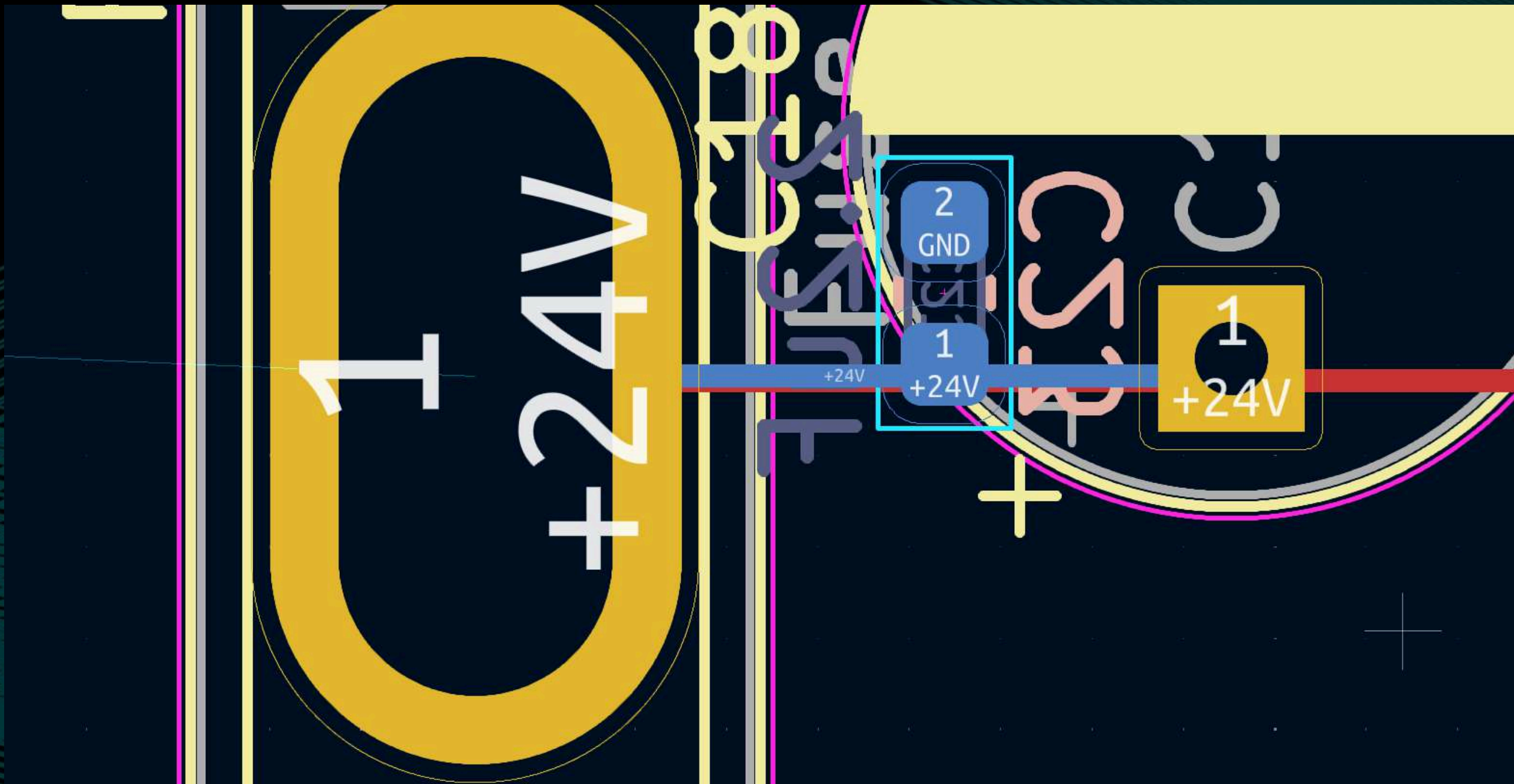
Do not put any track under the oscillator

Try not to put the oscillator in the back (your teammate may break it 😊)

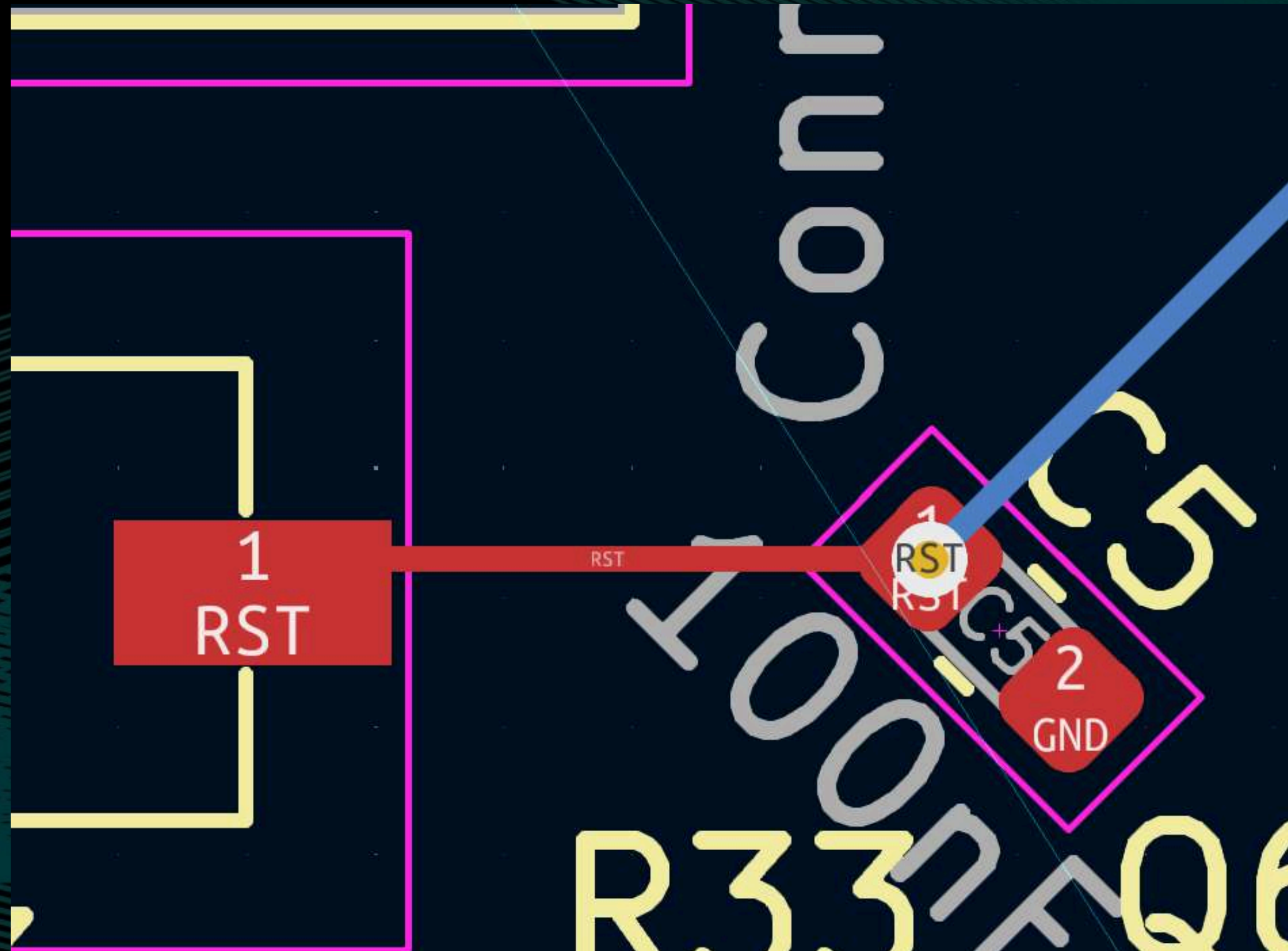
Remainder: try to put the MCU and the oscillator on the same side
and the track shd be same length



Key point you need to prevent:
Do not use thin track for high voltage / power wire
Otherwise the track may exploded

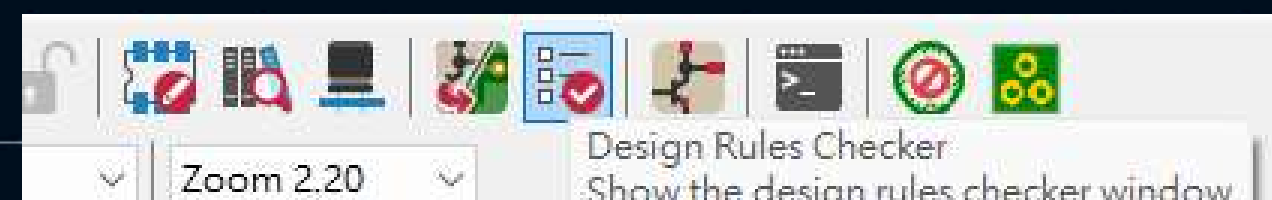


Key point you need to prevent:
Try to prevent putting via right on the pad(if you plan to
smd)

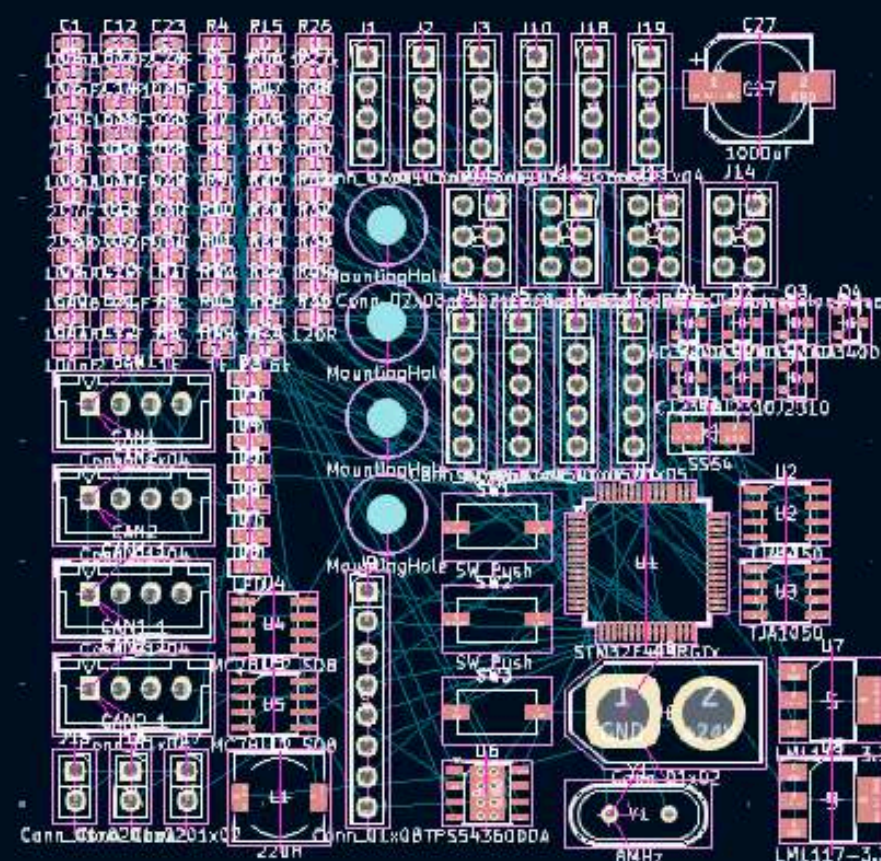




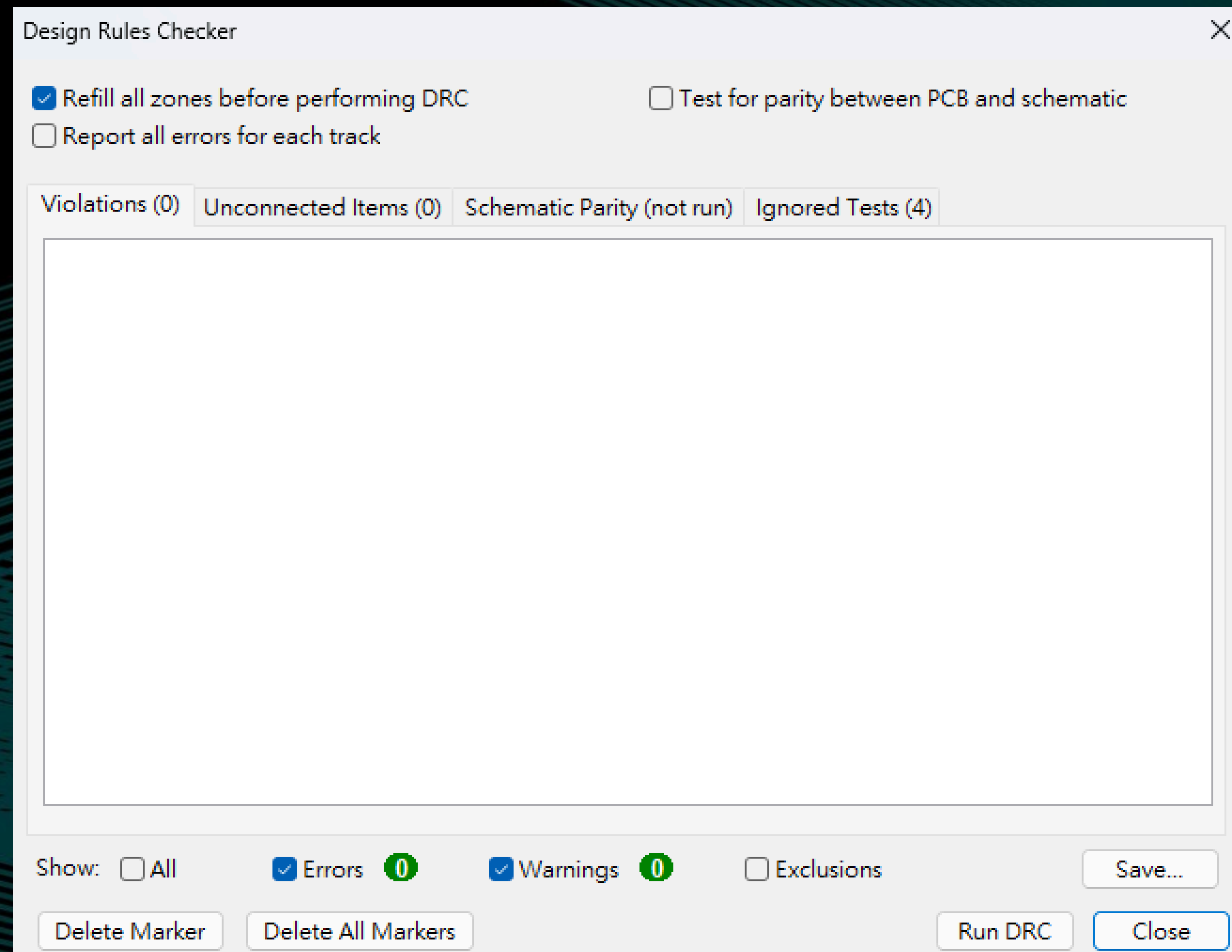
Track: use netclass width Via: use netclass sizes F.Silkscreen 0.0100 mm (0.39 mils) Zoom 1.50



Remember to run the DRC , vias stitching



Make sure to submit the PCB homework after 0 error in DRC



PCB

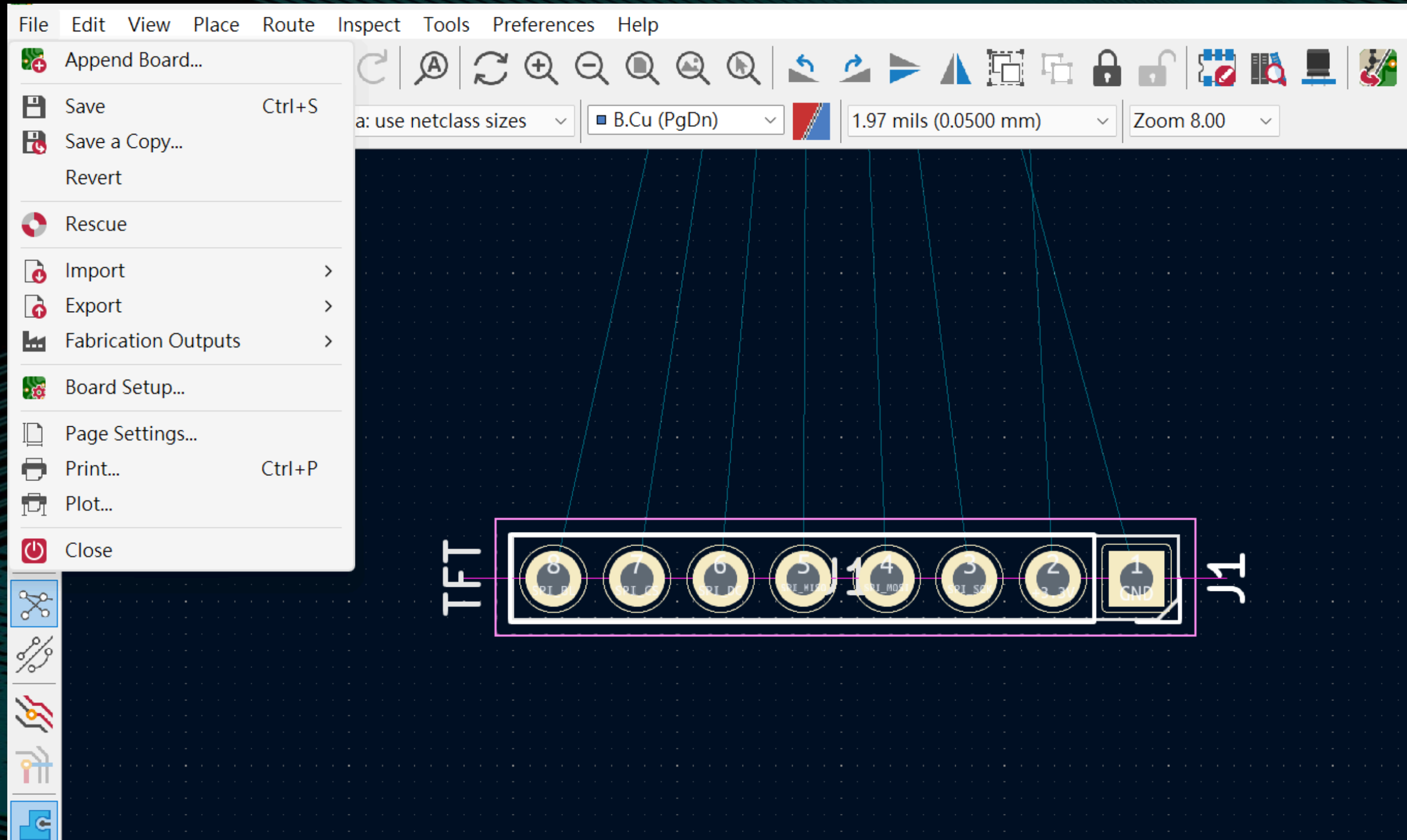
**LET'S HAVE A HOMEWORK OF TURNING SCHEMATIC TO A
PCB**

The Deadline is one week after tutorial 2

PCB

How to import TFT graphics

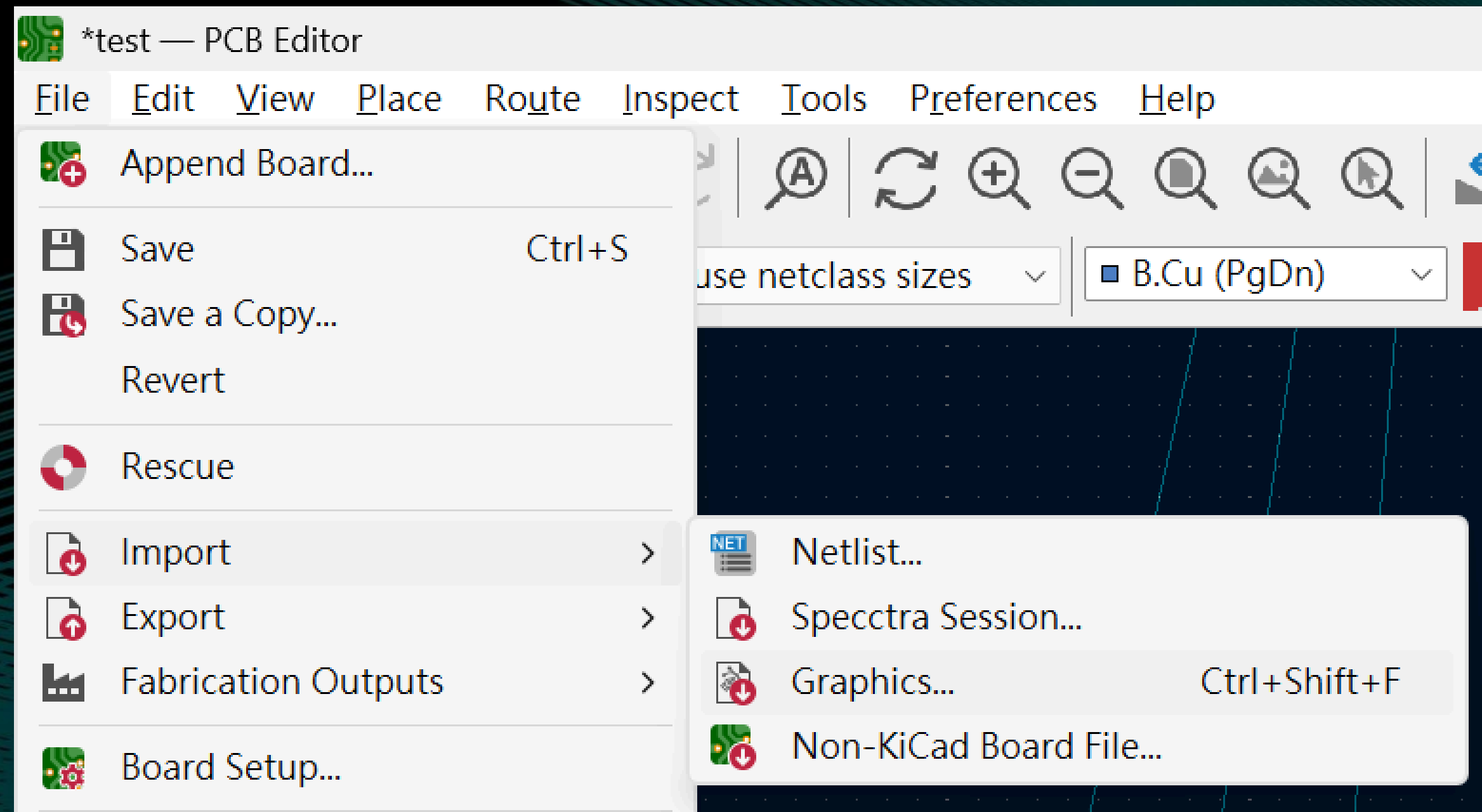
Press file



PCB

How to import TFT graphics

Import -> Graphics



PCB

How to import TFT graphics

Press this



Import Vector Graphics File

File: 

Import scale:

DXF default line width: mils

DXF default units:

☐ Place at: X: Y: mils

☒ Layer:

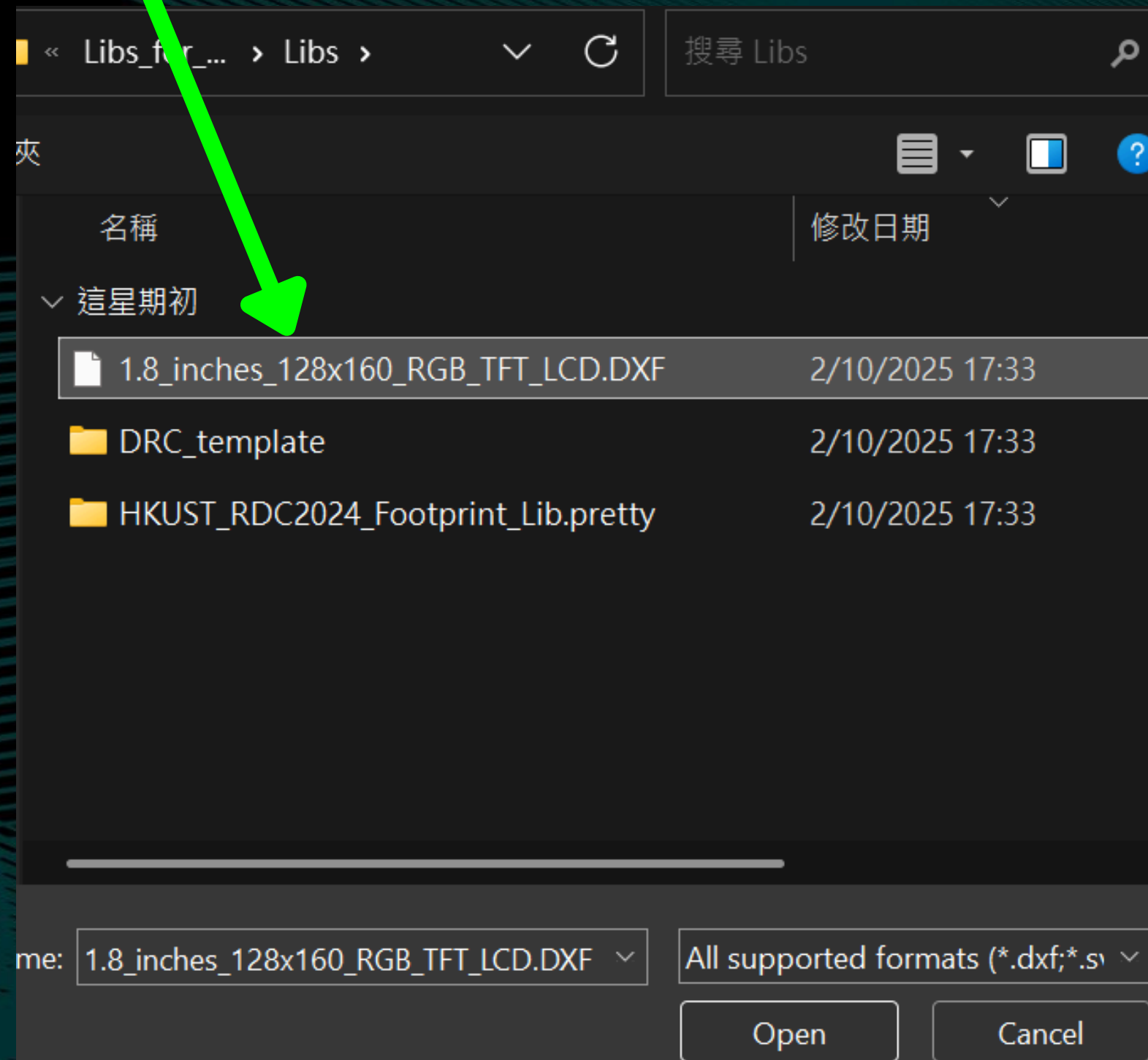
☒ Group imported items

☒ Fix discontinuities Tolerance: mils

PCB

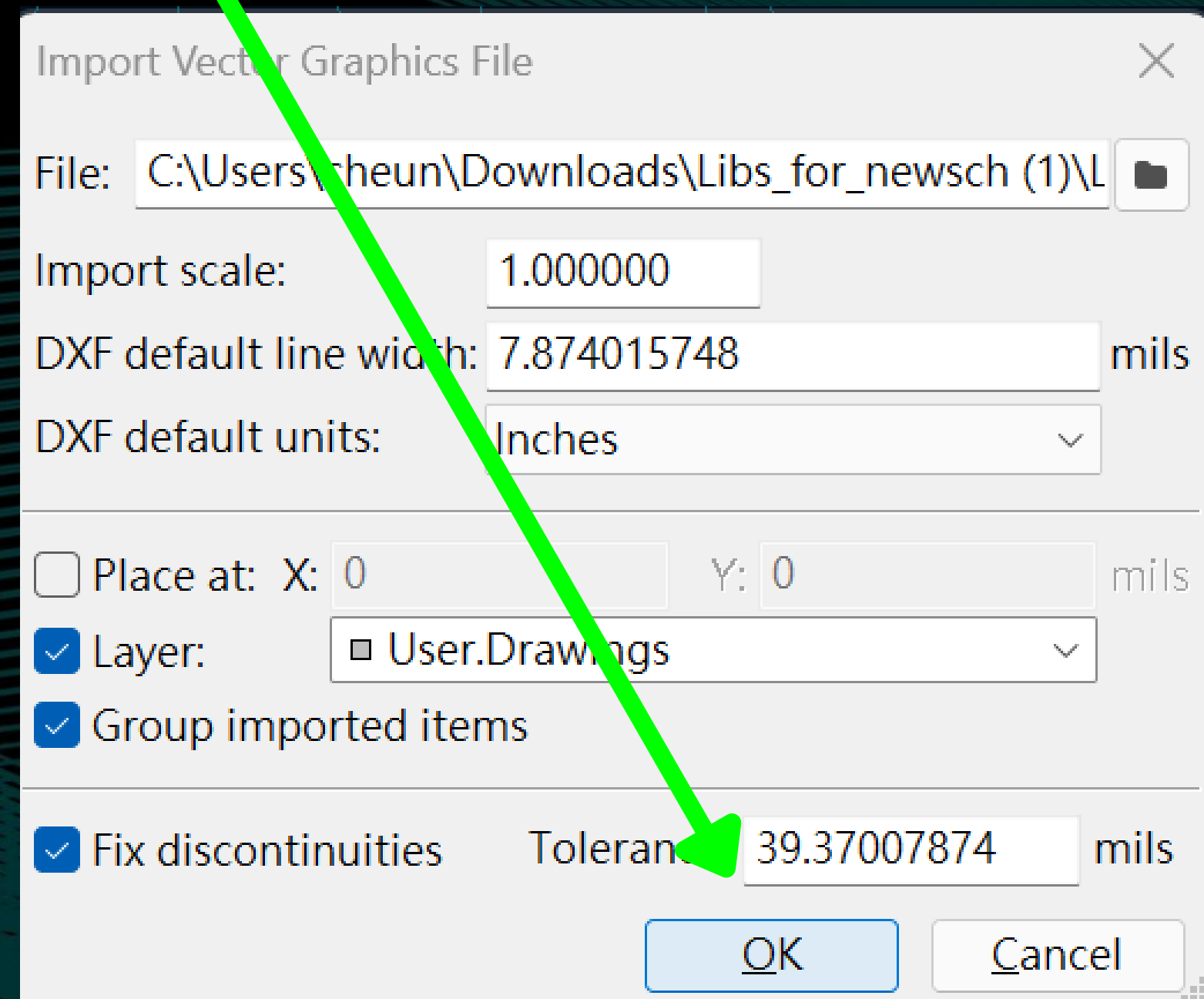
How to import TFT graphics

Find this file in the library file and open it



How to import TFT graphics OK!!!!

PCB



Import Vector Graphics File

File: C:\Users\cheun\Downloads\Libs_for_newsch (1)\L

Import scale: 1.000000

DXF default line width: 7.874015748 mils

DXF default units: Inches

☐ Place at: X: 0 Y: 0 mils

☒ Layer: User.Drawings

☒ Group imported items

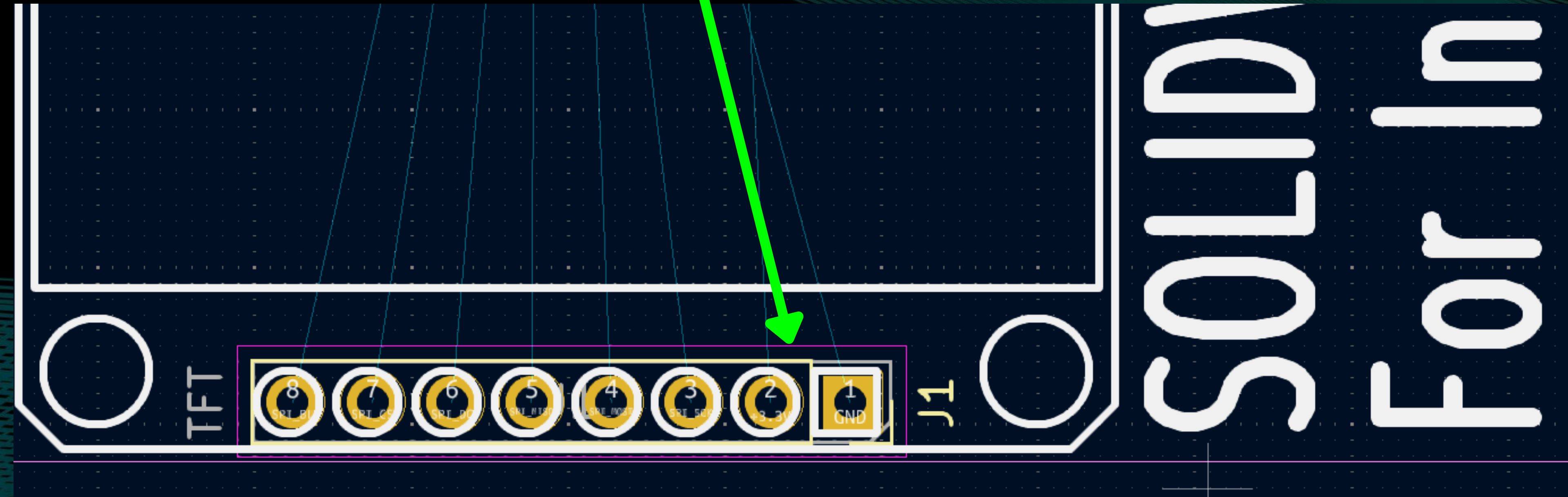
☒ Fix discontinuities Tolerance: 39.37007874 mils

OK Cancel

A green arrow points from the top left towards the 'Tolerance' field in the 'Fix discontinuities' section.

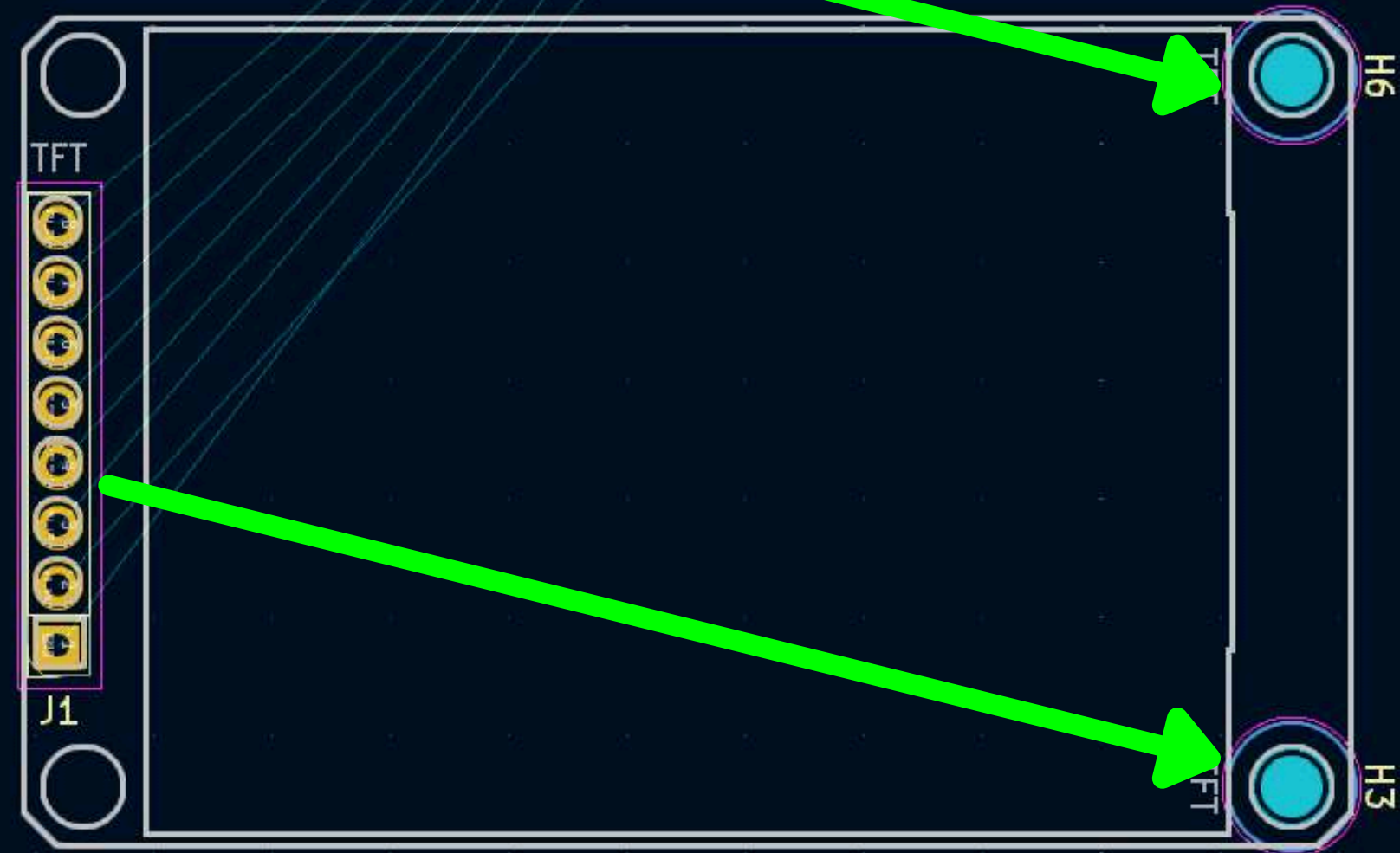
How to import TFT graphics
Align the graphics like this, around is fine
But the orientation must be correct

PCB



How to import TFT graphics
Align the mounting holes like this

PCB

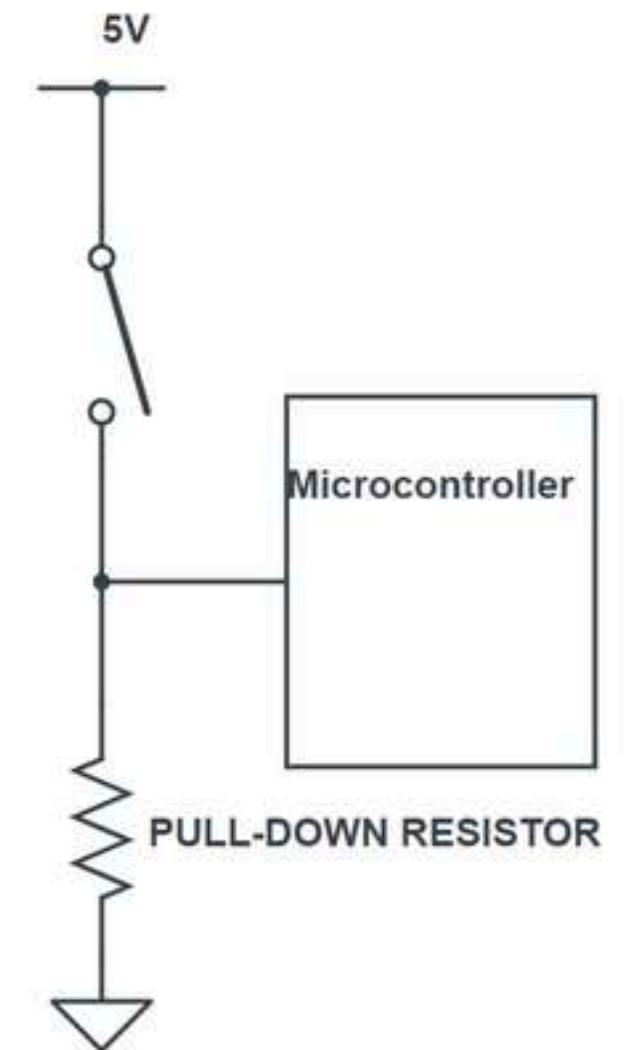
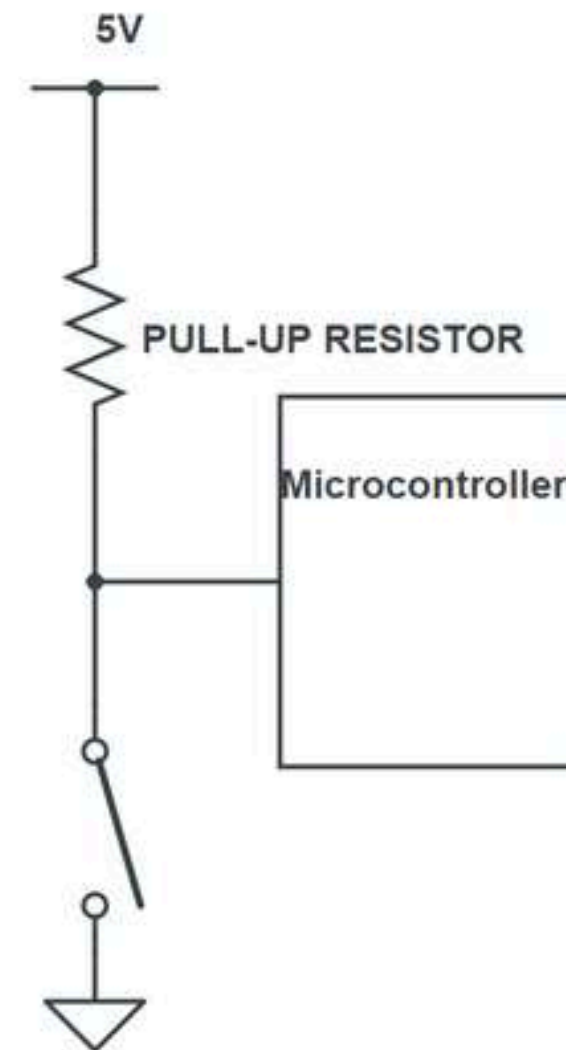


SOLIDWORKS Educational
For Instructional Use Only

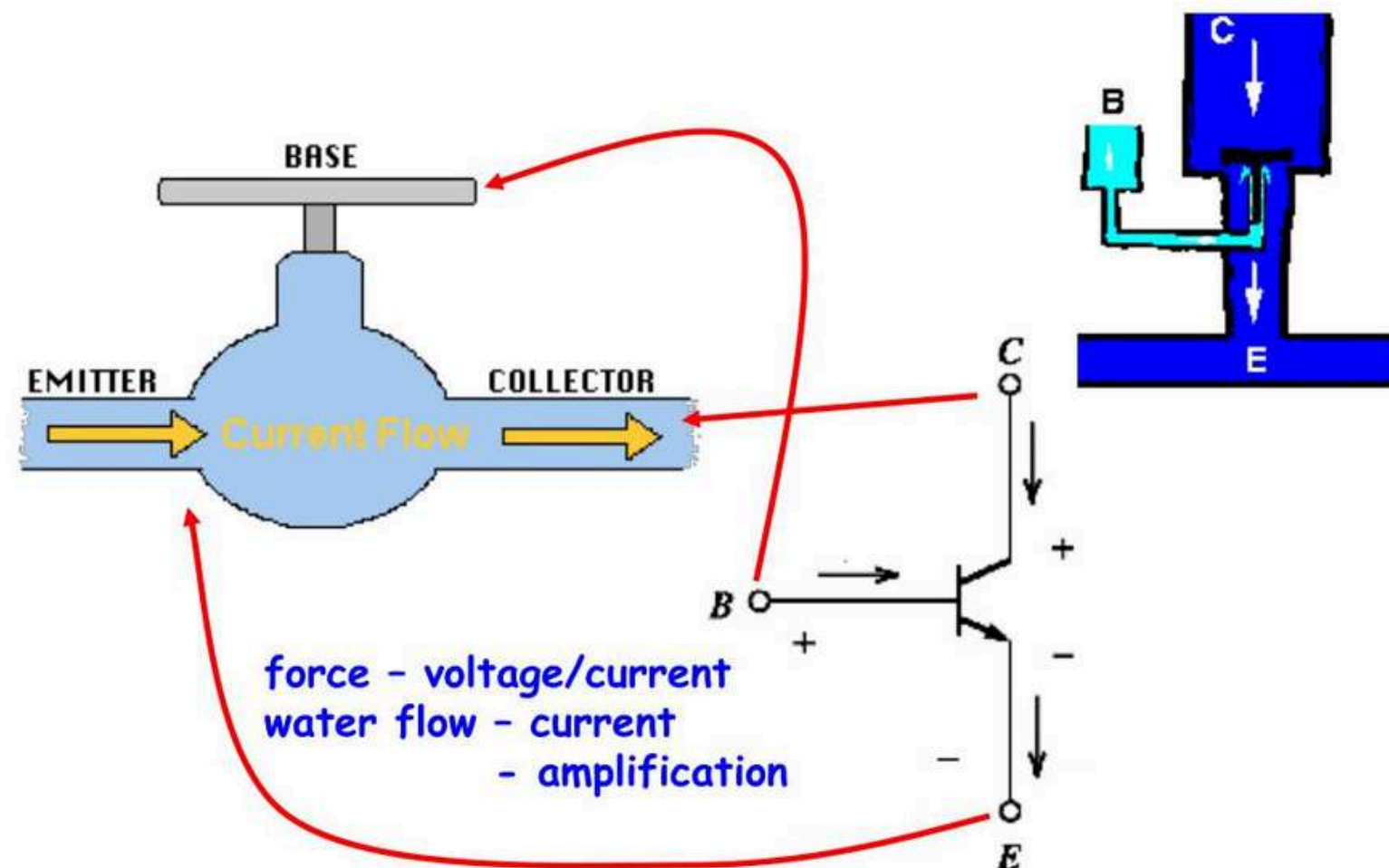
**Some concept
and theory stuff**

Pull-up / Pull-down

- Pull-up resistor



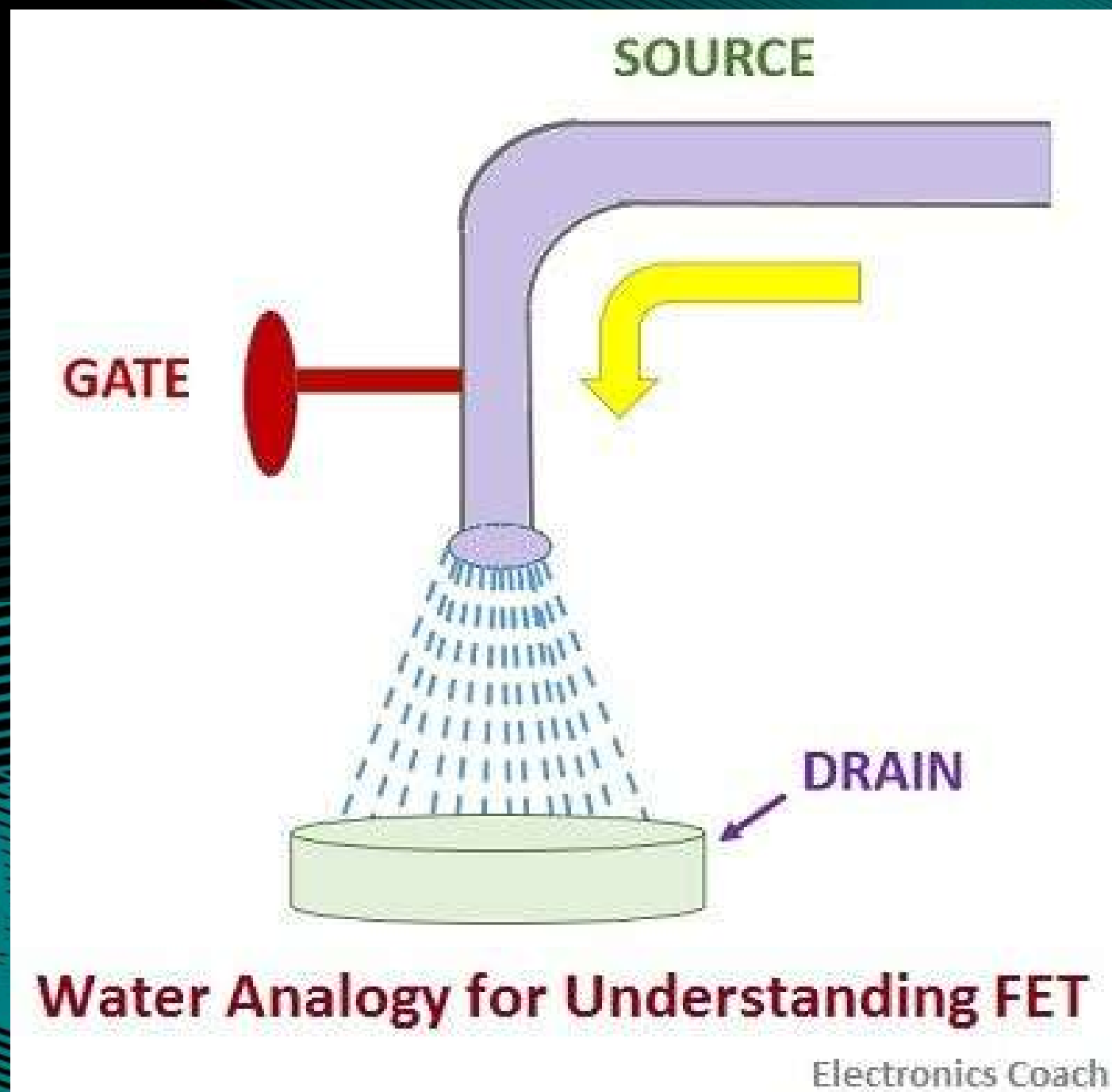
Bipolar Junction Transistor (BJT)



- BJT is an electronic switch
 - Allow to use small current to control high current
 - Use GPIO/PWM to control high current for different purpose
- NPN
 - Give small current to the base for allowing
 - High enable
 - current flow from collector to emitter
- PNP
 - Give small current to the base for allowing
 - Low enable
 - current flow from emitter to collector

MOSFET

- MOSFET is an electronic switch
 - Allow to use small current to control high current
 - Use GPIO/PWM to control high current for different purpose
- N-Channel MOSFET
 - Give positive voltage to the gate for allowing current flow from drain to source
- P-Channel MOSFET
 - Give negative voltage to the gate for allowing current flow from source to drain
- N-Channel MOSFET has higher efficiency than P-Channel MOSFET
 - N-Channel MOSFET is preferred



But what is the difference between MOSFET and BJT while they are all electronic switch

BJT(Bipolar Junction Transistor). (supplementary reading)

BJT (Bipolar Junction Transistor)

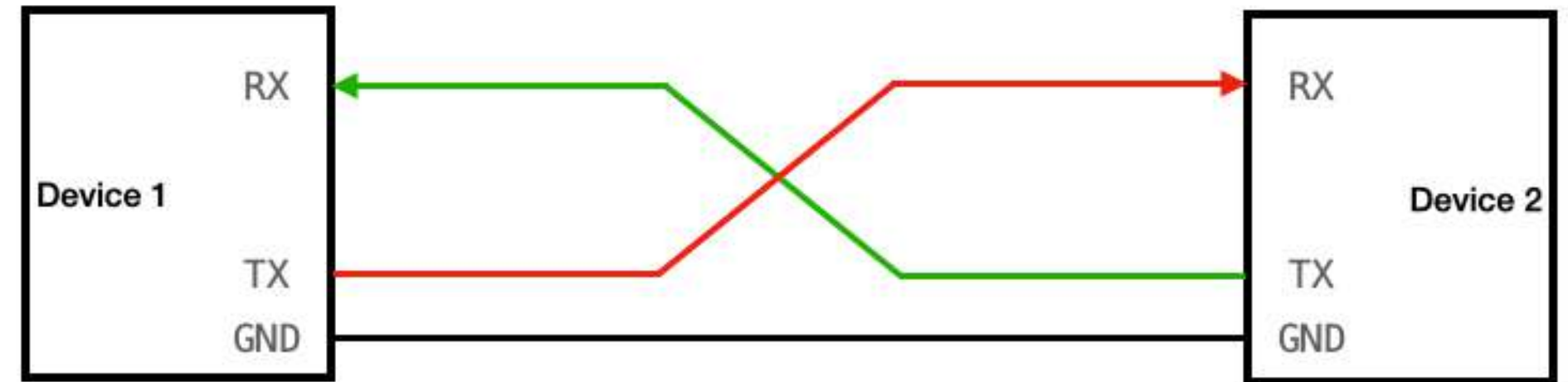
Advantages	Disadvantages
Can make small signals stronger for further use	Needs constant current to stay on, which uses more energy
Works well in circuits that need more current	Not as fast as MOSFETs at turning on and off
Gives steady and reliable flow of current	Can get hot, especially with heavy use
Ideal for signals that change continuously	Takes up more space in circuit boards compared to MOSFETs
A tiny current at the base controls a bigger current flow	May struggle with high-speed on/off tasks

MOSFET (supplementary reading)

MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor)	
Advantages	Disadvantages
Uses very little power when switching on and off	Can be damaged by static electricity if not handled carefully
Turns on and off quickly for high-speed tasks	Reacts badly to sudden voltage changes or spikes
Wastes less energy, so it generates less heat	May cost more than BJTs in some cases
Needs almost no current—only voltage at the gate	May not handle as much current as a BJT in similar conditions
Fits easily into small and modern electronic devices	Requires proper wiring and protection in power circuits

UART(Universal Asynchronous Receiver/Transmitter)

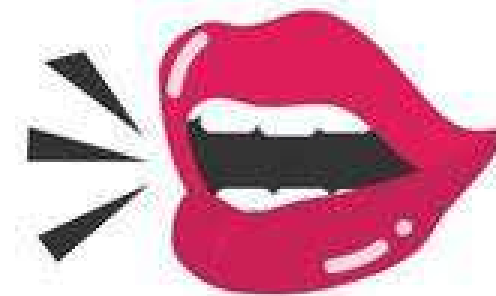
- UART
 - Serial transmission/reception,



TX



TX



RX

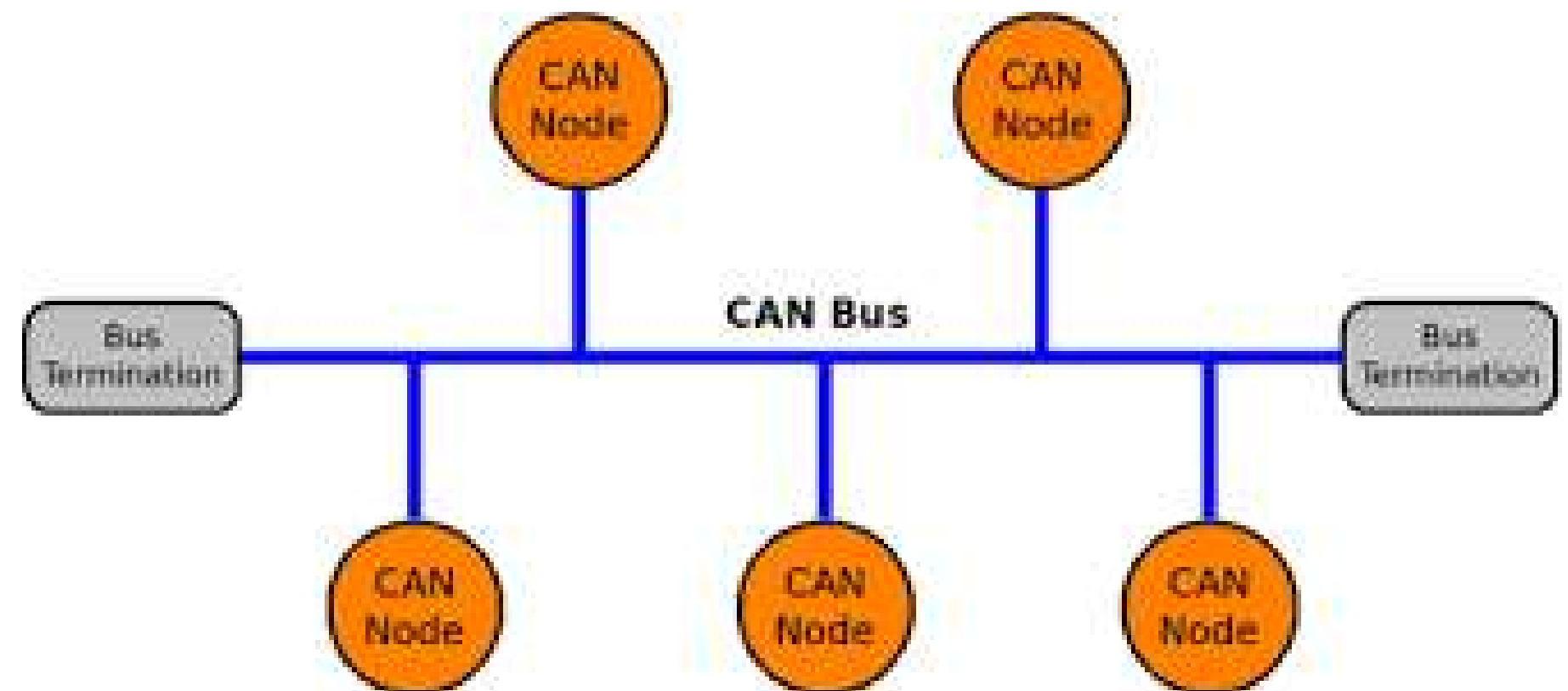


RX



CAN

- Controller Area Network (CAN)
 - Allow to control more than one device with the same connection bus



Serial Wire Debug (SWD) protocol

- Software will use ST Link
 - Flash code
 - Debug
- SWDIO
 - For data input and output
- SWCLK
 - For clock signal

